

Article

Numerical Study of the Basic Finner Model in Rolling Motion [†]

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[†] This paper is an extended version of our paper published in Ionut Bunescu, Mihai-Vladut Hothazie, Mihai-Victor Pricop, Alexandru Onel and Tudorel Afilipoae. Numerical Investigation of Basic Finner Model in Roll Motion as Complement to the Experimental Work. In Proceedings of the Conference Paper AIAA 2024–2348 at the AIAA SciTech Forum, Orlando, FL, USA, 8–12 January 2024.

Abstract: A numerical investigation of the roll motion characteristics of the Basic Finner Model was performed. The study of roll motion is essential in the design and performance evaluation of aerospace vehicles, particularly for stability and maneuverability purposes. The numerical investigation was conducted employing the Unsteady Reynolds-Averaged Navier-Stokes (URANS) solver coupled with $k-\epsilon$ realizable turbulence model. The simulations were performed for a range of Mach numbers and angles of attack to assess the influence of these parameters on the model’s roll motion characteristics. The CFD procedure was validated based on an experimental database from previous work and the literature. The influence of roll motion on aerodynamic forces and moments at different flow conditions were analyzed to obtain a better understanding of the physics. The variation of forces and moments with roll angle, Mach number, and angle of attack, as well as the pressure distribution at different flow conditions, are discussed, also covering aerodynamic interactions between the fins and body. This numerical investigation contributes to understanding the aerodynamic behavior of the Basic Finner Model during roll motion. The findings are valuable for the design and optimization of aerospace vehicles, aiding in the development of more efficient and stable configurations. Future research can be based upon these results to explore additional factors that may impact roll motion characteristics and can further refine the design and performance evaluation processes for aerospace vehicles.

Keywords: unsteady flow simulation; CFD; roll damping coefficient



Academic Editor: Anthony D. Gardner

Received: 25 February 2025

Revised: 23 April 2025

Accepted: 23 April 2025

Published: 24 April 2025

Citation: Bunescu, I.; Hothazie, M.-V.; Stoican, M.-G.; Pricop, M.-V.; Onel, A.-I.; Afilipoae, T.-P. Numerical Study of the Basic Finner Model in Rolling Motion. *Aerospace* **2025**, *12*, 371. <https://doi.org/10.3390/aerospace12050371>

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1. Introduction

The evaluation of aerodynamic coefficients for a model in motion requires either a dynamic experimental approach or numerical analysis. Experimental testing in wind tunnels can be conducted relatively quickly, once the experimental setup is available; however, this requires specialized facilities and is influenced by various factors such as wall effects, model support structures (sting), model cavity disturbances, or slight geometric deviations.

Conversely, numerical analysis presents a viable alternative, as it enables the simulation of flow across a broad range of Mach numbers, incorporates turbulence modeling, and does not require extensive experimental infrastructure. However, numerical simulations,

particularly for unsteady flow conditions, demand significant computational resources and extended processing times.

The present study examines the influence of roll motion on the aerodynamic coefficients of a slender body over a range of Mach numbers, angles of attack, and roll angles, extending the analysis introduced in [1]. Building upon the numerical investigations presented in [2–4], this research aims to further characterize the aerodynamic behavior of a slender body undergoing rolling motion. Additionally, it extends the analysis conducted in [5–7] by exploring variations in aerodynamic coefficients and derivatives as a function of key flow and motion parameters.

In the context of rolling motion, several studies have investigated the aerodynamic characteristics of standard models, such as the Basic Finner Model and the Modified Basic Finner Model, under different flow conditions. The study presented in [5] analyzed these models at Mach 2.49 and angles of attack ranging from -5° to 90° using Unsteady Reynolds-Averaged Navier-Stokes (URANS) simulations with a $k-\epsilon$ turbulence model. This paper [5] presented studies on the dependence of grid resolution, physical time step, and the number of inner iterations to accurately replicate the experimental results obtained in [8].

Further numerical investigations of the Modified Basic Finner Model were conducted in [6], where quasi-stationary and unsteady approaches were employed at Mach 1.15 and 1.75 using the Reynolds-Averaged Navier-Stokes (RANS) equations with a $k-\omega$ SST turbulence model. The numerical results were compared with experimental data from [9,10]. Both approaches presented good results for roll damping coefficient variation with angle of incidence.

A comprehensive analysis comparing unsteady and quasi-steady approaches was presented in [3], where numerical simulations were performed on the Basic Finner Model at Mach 0.4 and Mach 2.5 for angles of attack of 0° , 10° , and 20° . The findings indicated that, while both approaches present satisfactory results, the unsteady model provides higher accuracy at the cost of significantly increased computational resources. In contrast, the quasi-steady model requires significantly less computational effort, making it a practical choice for certain applications, though it comes at the cost of reduced accuracy.

Despite these contributions, numerical studies on rolling motion for standard models have been relatively limited in scope. Many analyses focus either on a single Mach number with multiple angles of attack, as in [5], or on a single angle of attack across multiple Mach numbers, as in [11]. Given the increasing need for accurate predictions of dynamic stability in aerospace applications, further research is required to examine a broader range of flow conditions. While some previous studies have primarily presented experimental findings [9,12–15], others have offered limited numerical results [2,5,6].

The present study aimed to contribute to these research field by conducting a comprehensive numerical analysis of the Basic Finner Model in rolling motion. The investigation covered multiple Mach regimes, including subsonic (Mach 0.4), transonic (Mach 0.95), and supersonic (Mach 1.6, 2.5, and 3.5), with angles of attack varying from 0° to 50° . Furthermore, the aerodynamic coefficients were analyzed over a full 360° roll angle rotation to capture the effects of rolling motion comprehensively.

Given their sensitivity to rolling motion, key aerodynamic coefficients of interest include the roll damping moment coefficient (C_{l_p}), yaw moment spin derivative coefficient (C_{n_p}), and side force spin derivative coefficient (C_{Y_p}). Additionally, other relevant aerodynamic coefficients, such as the pitch moment coefficient (C_m), normal force coefficient (C_N), axial force coefficient (C_X), lift coefficient (C_L), and drag coefficient (C_D), were examined as functions of the considered parameters.

This study's complexity was defined by the interaction of detached flow phenomena, shock-wave effects, and vortex structures, combined with rolling motion. To ensure the

reliability of the numerical results, selected cases were validated against available experimental data. Ultimately, the findings aim to enhance understanding of roll behavior in finned projectiles and contribute to the development of more effective and controllable guided missile systems.

2. Study Setup

2.1. Background

The aerodynamic coefficients studied in this research are presented in Table 1.

Table 1. Definition of aerodynamic coefficients.

Roll damping coefficient	$C_{l_p} = \frac{M_L}{\frac{1}{2}\rho V^2 S d \left(\frac{pd}{2V}\right)}$
Yaw moment spin derivative coefficient	$C_{n_p} = \frac{M_N}{\frac{1}{2}\rho V^2 S d \left(\frac{pd}{2V}\right)}$
Side force spin derivative coefficient	$C_{Y_p} = \frac{Y}{\frac{1}{2}\rho V^2 S \left(\frac{pd}{2V}\right)}$
Pitch moment coefficient	$C_m = \frac{M_M}{\frac{1}{2}\rho V^2 S d}$
Normal force coefficient	$C_N = \frac{N}{\frac{1}{2}\rho V^2 S}$
Axial force coefficient	$C_X = \frac{X}{\frac{1}{2}\rho V^2 S}$
Lift coefficient	$C_L = \frac{L}{\frac{1}{2}\rho V^2 S}$
Drag coefficient	$C_D = \frac{D}{\frac{1}{2}\rho V^2 S}$

Where M_L , M_M , and M_N are the aerodynamic roll, pitch, and yaw moments; Y , N , and X are the aerodynamic side force, normal force, axial force, lift force, and drag force; ρ is the air density; V is the air speed; S is the reference area $\left(\frac{\pi d^2}{4}\right)$, and d is the model reference diameter.

The aerodynamic forces (X , Y , N) and moments (M_L , M_M , and M_N) were evaluated in the body-fixed coordinate system $Oxyz$, as presented in Figure 1.

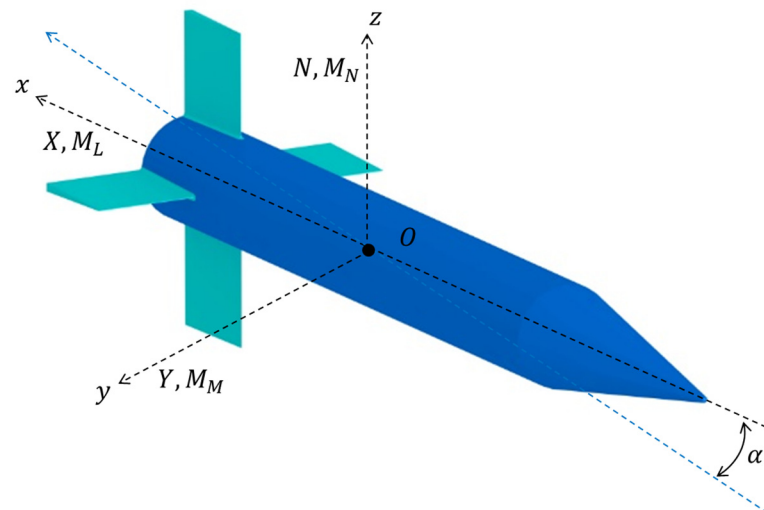


Figure 1. Axis system.

2.2. Considered Geometry

The model used to evaluate the aerodynamic characteristics of rolling motion under varying parameters was the Basic Finner Model, also known as the Army-Navy Finner [9]. This model consists of a cone-cylinder body with four rectangular fins, as shown in Figure 2. This model was selected due to its recognition by the Supersonic Tunnels Association International (STAI) and the Advisory Group for Aerospace Research and Development

(AGARD) as the standard for dynamic testing, having been extensively used in wind tunnel experiments and calibration procedures [15]. The studied model measures 60 mm in diameter and 600 mm in length. The reference point of the model was placed at 6.1d with respect to the model's nose, similar to [8].

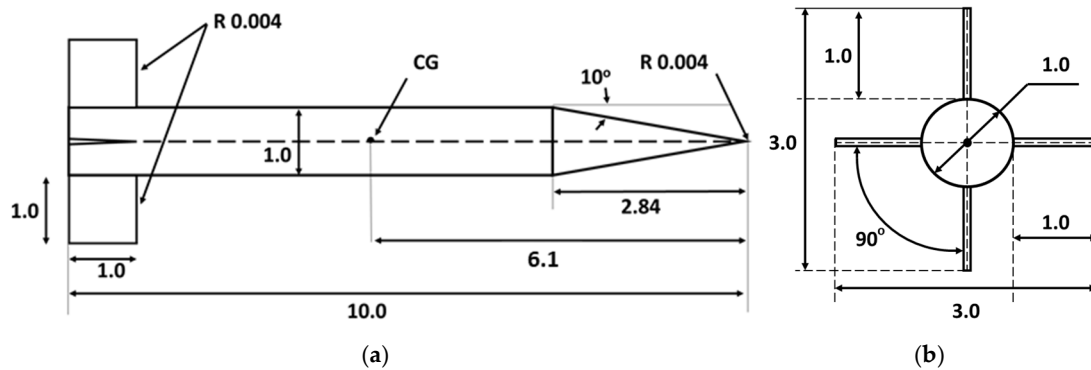


Figure 2. Basic finner geometry: (a) side view; (b) front view.

2.3. Mathematical Modeling

The numerical investigation was conducted using the unsteady Reynolds-averaged Navier-Stokes (URANS) model, coupled with the realizable k - ϵ turbulence model [16]. These models have been shown to provide accurate results for rotating flow problems, as evidenced in [5], while offering a lower computational cost compared to alternative approaches such as LES, DES, or URANS with various k - ω turbulence models, primarily due to the less demanding requirements for the first cell height near the wall.

The governing equations of the RANS model are presented below. Equation (1) corresponds to the mass conservation equation, Equation (2) represents the momentum conservation equations along the x , y , and z axes, and Equation (3) describes the energy conservation equation.

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u_i)}{\partial x_i} = 0, \quad (1)$$

$$\frac{\partial (\rho u_i)}{\partial t} + \frac{\partial}{\partial x_j} (\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} (\tilde{\tau}_{ij} - \rho \langle u'_i u'_j \rangle), \quad (2)$$

$$\frac{\partial (\rho E)}{\partial t} + \frac{\partial}{\partial x_j} [\rho u_i H] = \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + \frac{\partial}{\partial x_j} (\tilde{\tau}_{ij} u_i + \langle u'_i \tilde{\tau}_{ij} \rangle - \langle \rho u'_j H' \rangle), \quad (3)$$

where the term $\tilde{\tau}_{ij}$ represents the viscous stress tensor:

$$\tilde{\tau}_{ij} = \mu \left[\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right] + \mu \left[\left(\frac{\partial \langle u'_i \rangle}{\partial x_j} + \frac{\partial \langle u'_j \rangle}{\partial x_i} \right) - \frac{2}{3} \frac{\partial \langle u'_k \rangle}{\partial x_k} \delta_{ij} \right], \quad (4)$$

The terms in Equations (1)–(4) are defined in [17]: ρ represents the fluid density, u_i represents the fluid velocity along i -axis, p is the fluid pressure, E refers to the fluid internal energy, H is the fluid enthalpy, k is the thermal conductivity of the fluid, T is the temperature of the fluid, and δ_{ij} represents the Kronecker delta symbol.

The k - ϵ realizable model consists of Equations (5) and (6), which represent the transport equations for turbulent kinetic energy (k) and dissipation rate (ϵ).

$$\frac{\partial}{\partial t} (\rho k) + \frac{\partial}{\partial x_j} (\rho k u_j) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \epsilon - Y_M + S_k, \quad (5)$$

$$\frac{\partial}{\partial t}(\rho\varepsilon) + \frac{\partial}{\partial x_j}(\rho\varepsilon u_j) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + \rho C_1 S_\varepsilon - \rho C_2 \frac{\varepsilon^2}{k + \sqrt{\nu \varepsilon}} + C_{1\varepsilon} \frac{\varepsilon}{k} C_{3\varepsilon} G_b + S_\varepsilon, \quad (6)$$

where G_k and G_b represent turbulent kinetic energy production due to velocity gradients and floatability phenomena, Y_M represents the contributions of fluctuations of dilatation in compressible turbulence, and S_k and S_ε represent the source terms for turbulent kinetic energy k and dissipation rate ε . Moreover, the coefficients C_1 , $C_{1\varepsilon}$, and $C_{3\varepsilon}$ represent constants of the model, as presented in [18].

Numerical simulation of the model in rolling motion was carried out using the Sliding Mesh Technique [18,19] which involves the rotation of an inner domain that includes the model and information exchange between the stationary domain and the rotating domain at the interface.

2.4. Domain Discretization

The computational domain consists of a spherical stationary domain with diameter $D_D = 100$ m and a cylindrical rotating domain with length $L_D = 0.75$ m and diameter $d_D = 0.25$ m, as presented in Figure 3. The interface between the two domains allows transfer of information while the rotating domain changes its position around the longitudinal axis. The length of the inner domain (L_D) is 150 mm greater than that of the model, while its diameter (d_D) exceeds the fin span by 0.07 m. The diameter of the outer domain (D_D) is 166 times larger than the length of the model, ensuring good results in both subsonic and supersonic flow regimes.

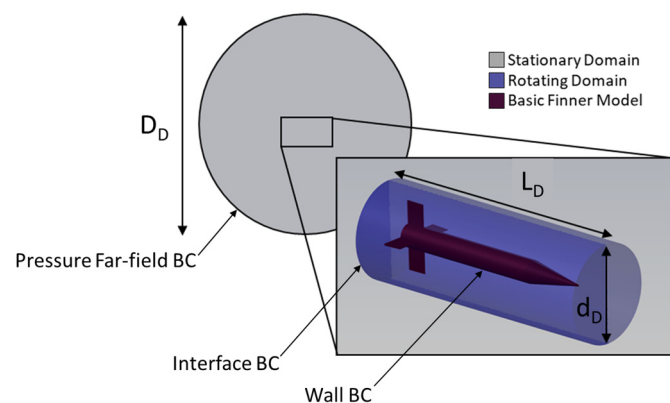


Figure 3. Computational domain.

The meshing of the two domains comprised 45 layers of prismatic cells in the vicinity of the model, with a growth factor of 1.1. The mesh also included cubic and polyhedral cells in both the rotating and stationary domains. The maximum mesh size on the model surface was 0.005 m, which allowed curvature and proximity refinements, as shown in Figure 4.

The cell size in the rotating domain ranged from 0.001 m and 0.01 m, while the maximum size in the stationary domain was 1 m, as can be seen in Figure 5, where the distribution of cell size on the vertical plane is presented. This mesh configuration was determined to be optimal following mesh convergence analysis. The first-layer thickness of the cells around the model varied with Mach number, such that y^+ was over 30. The final mesh consisted of around 9.6 million cells.

For the mesh convergence analysis, three grid levels were evaluated: a “coarse” grid (6.2 million cells), a “medium” grid (9.6 million cells), and a “fine” grid (12.4 million cells). Figure 6 illustrates the variation of ΔC_{l_p} with grid size for two Mach numbers (0.4 and 2.5) at an angle of attack (AoA) of 0° . Here, ΔC_{l_p} represents the difference between the C_{l_p} obtained from a given mesh and the C_{l_p} computed using the “fine” grid.

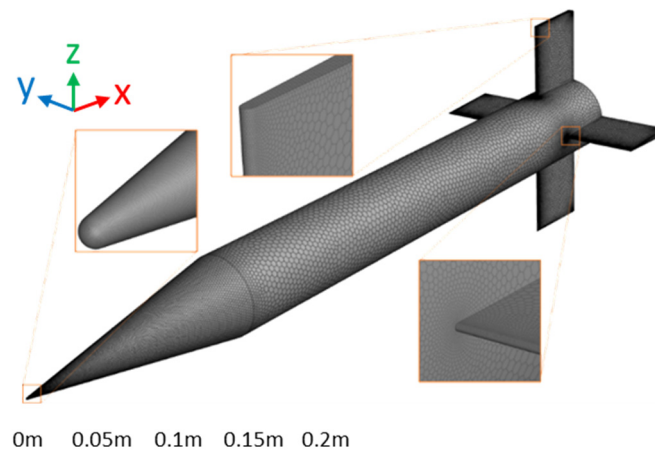


Figure 4. Grid size on the surface of the model.

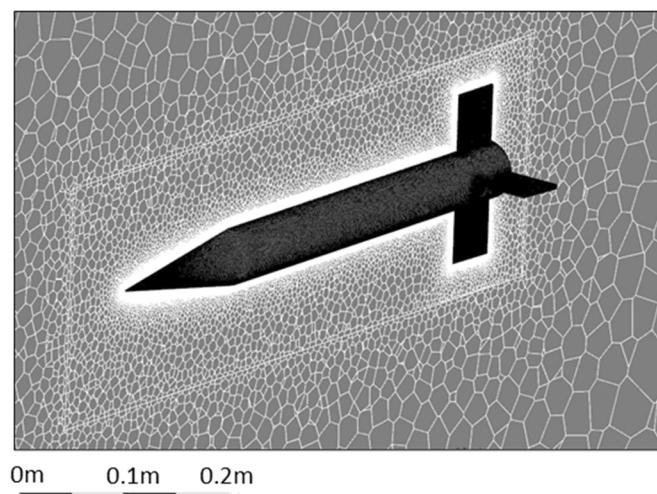


Figure 5. Grid size on the vertical plane.

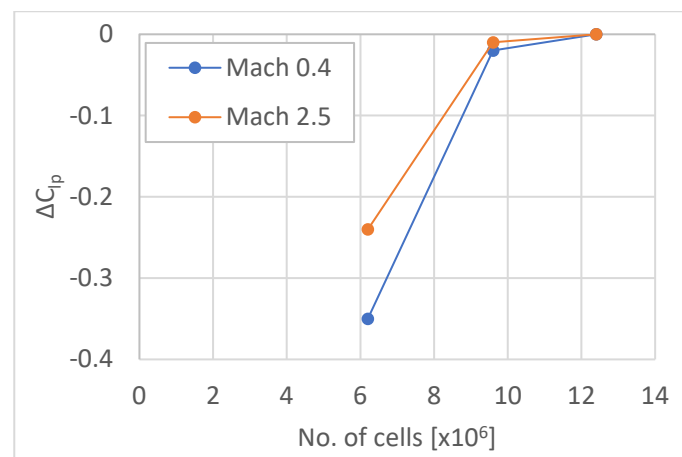


Figure 6. Mesh convergence analysis (C_{lp} averaged over roll angle).

Figure 6 shows that the “medium” grid was the optimal choice, as it provided a reduced number of cells while maintaining a minimal C_{lp} difference compared to the “fine” grid.

2.5. Solver Settings

The boundary conditions adopted were wall condition on the model surface, interface condition on the outer surface of the rotating domain and the inner surface of the stationary domain, and pressure far-field on the outer surface of the stationary domain, as presented in Figure 3. To solve the transient RANS equations, a second-order ROE scheme was used. The program used to perform this analysis was Fluent.

The spin parameter p^* , defined as $\frac{pd}{2V}$, was considered in this study to be equal to 0.005, while the time step was set with respect to angular step, which was 0.3° . The characteristic time varied from 1.31 s at Mach 0.4 to 0.27 s at Mach 3.5, so for a nondimensional time equal to $5 \cdot 10^{-5}$, the time step varied between $6.58 \cdot 10^{-5}$ s at Mach 0.4 to $1.37 \cdot 10^{-5}$ s at Mach 3.5. For each time step, 100 inner iterations were performed to obtain good convergence of the solution.

The characteristic time t_c represents the time it takes for a fluid particle to travel a characteristic length (i.e., model length) due to convection phenomena (i.e., flow velocity). It is defined as: $t_c = \frac{L_{ref}}{V_{ref}}$.

On the other hand, the non-dimensional time (also called the dimensionless time) is a scaled representation of time in a system, typically used to generalize results and compare different cases in fluid dynamics. It is defined as a ratio between the physical time t and characteristic time t_c : $t^* = \frac{t}{t_c}$.

In [5] it was shown that a non-dimensional time of approximately 10^{-5} is highly suitable for rotating applications. Additionally, the paper presented an analysis of the dependence of dynamic coefficients on the physical timestep, concluding that a non-dimensional time of approximately 10^{-5} is very effective.

3. Results

3.1. CFD Validation with Experimental Data

The first set of results compare the accuracy of CFD solutions with experimental ones obtained in different wind tunnels: Naval Ordnance Laboratory (NOL) [13], Ballistic Research Laboratory (BRL) [12], National Aerospace Laboratories (NAL) [15], and National Institute for Aerospace Research (INCAS) [2]. Figure 7 presents the variation in the roll damping moment coefficient at $\text{AoA} = 0^\circ$. Overall, the numerical results were in strong agreement with the experimental data, with the exception of the result at Mach 1.6, where the numerical simulation overestimated the experimental roll damping coefficient.

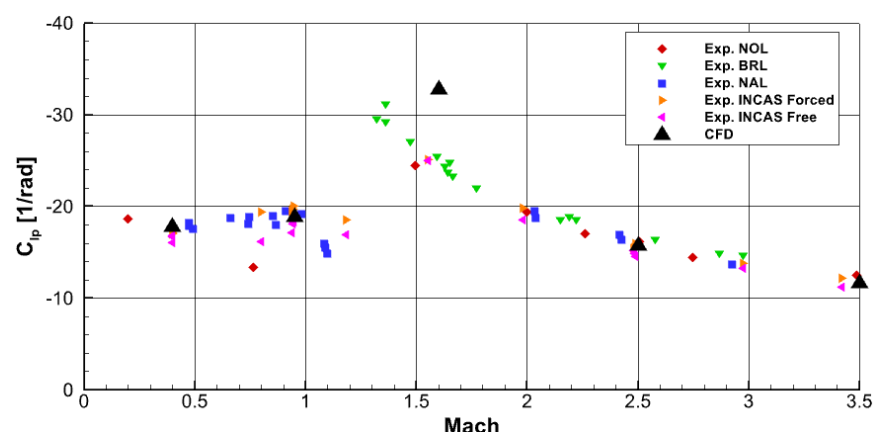


Figure 7. CFD results validation for various Mach numbers at $\text{AoA} = 0^\circ$ (C_{l_p} averaged over roll angle).

For the variation of the angle of attack, a single experimental dataset is available at Mach 2.5 [9]. Therefore, Figure 8 presents the variation in aerodynamic coefficients with angle of attack to compare the accuracy of the CFD results with the experimental data.

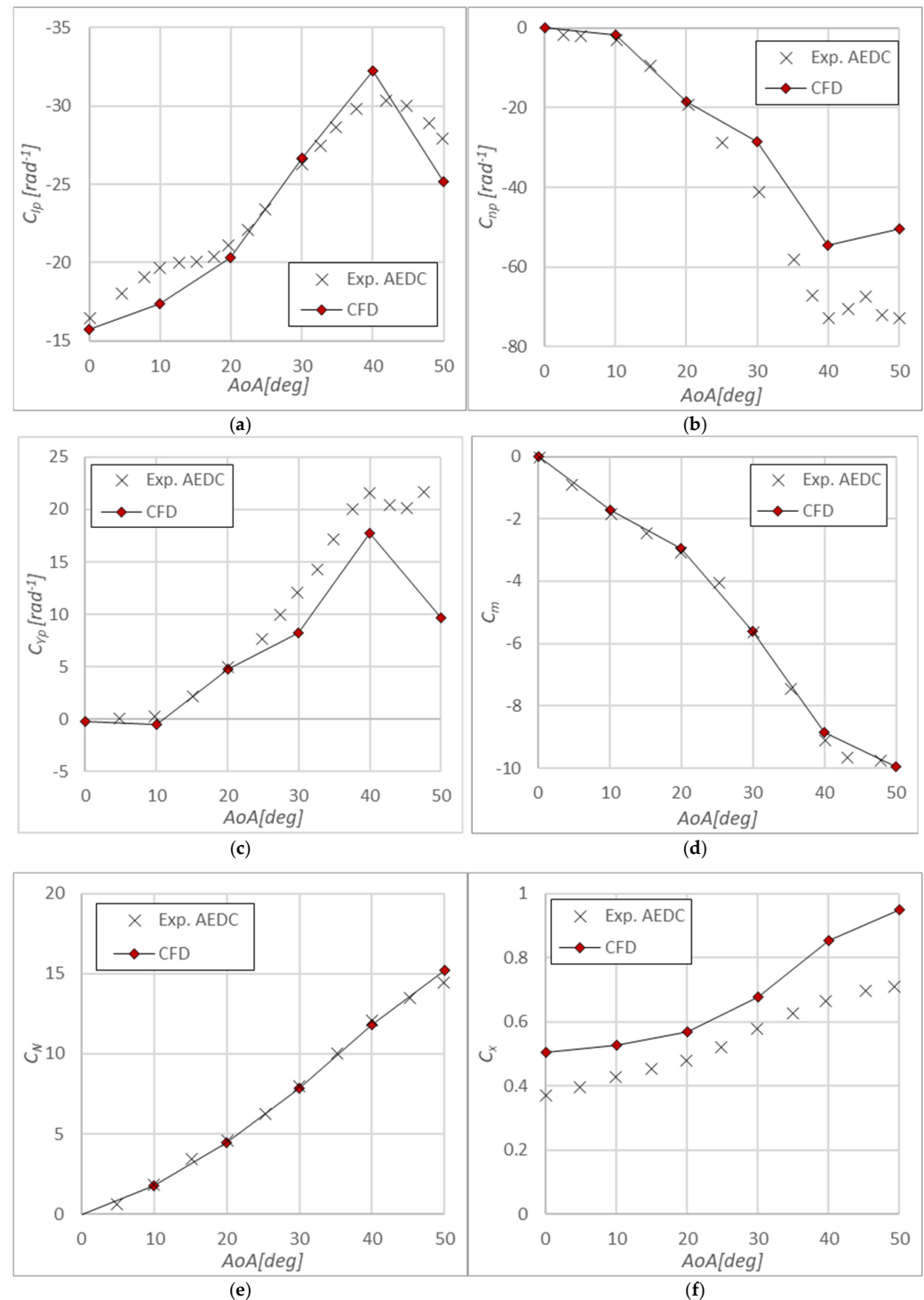


Figure 8. Validation of CFD settings for Mach 2.5: (a) C_{l_p} vs. AoA validation, (b) C_{n_p} vs. AoA validation, (c) C_{Y_p} vs. AoA validation, (d) C_m vs. AoA validation, (e) C_N vs. AoA validation, and (f) C_X vs. AoA validation; all data are averaged over roll angle.

All aerodynamic coefficients presented a consistent overall trend that aligned with the experimental data. However, it can be observed that C_{l_p} variation (Figure 8a), C_{n_p} variation (Figure 8b), C_{Y_p} variation (Figure 8c), and C_m variation (Figure 8d) demonstrated higher accuracy at low angles of attack, while slight deviations were noted at higher incidences. As reported in [5], the CFD results for C_{n_p} and C_{Y_p} at incidence angles greater than 30° presented slightly lower absolute values compared to the experimental data.

The variation in C_N (Figure 8e) was in line with the experimental data, while the C_X variation (Figure 8f) followed a consistent trend but was overpredicted by approximately 25%. This can be attributed to factors such as the sting interference, cavity pressure, and base drag contribution. However, the CFD predictions demonstrated high overall accuracy, making them suitable for qualitative analysis of the aerodynamic characteristics of the Basic Finner Model in rolling motion.

3.2. Aerodynamic Coefficients Variation with Angle of Attack

Figure 9 presents the variations in C_{l_p} (Figure 9a), C_{n_p} (Figure 9b), C_{Y_p} (Figure 9c), C_m (Figure 9d), C_N (Figure 9e), and C_X (Figure 9f) with angle of attack for various Mach numbers.

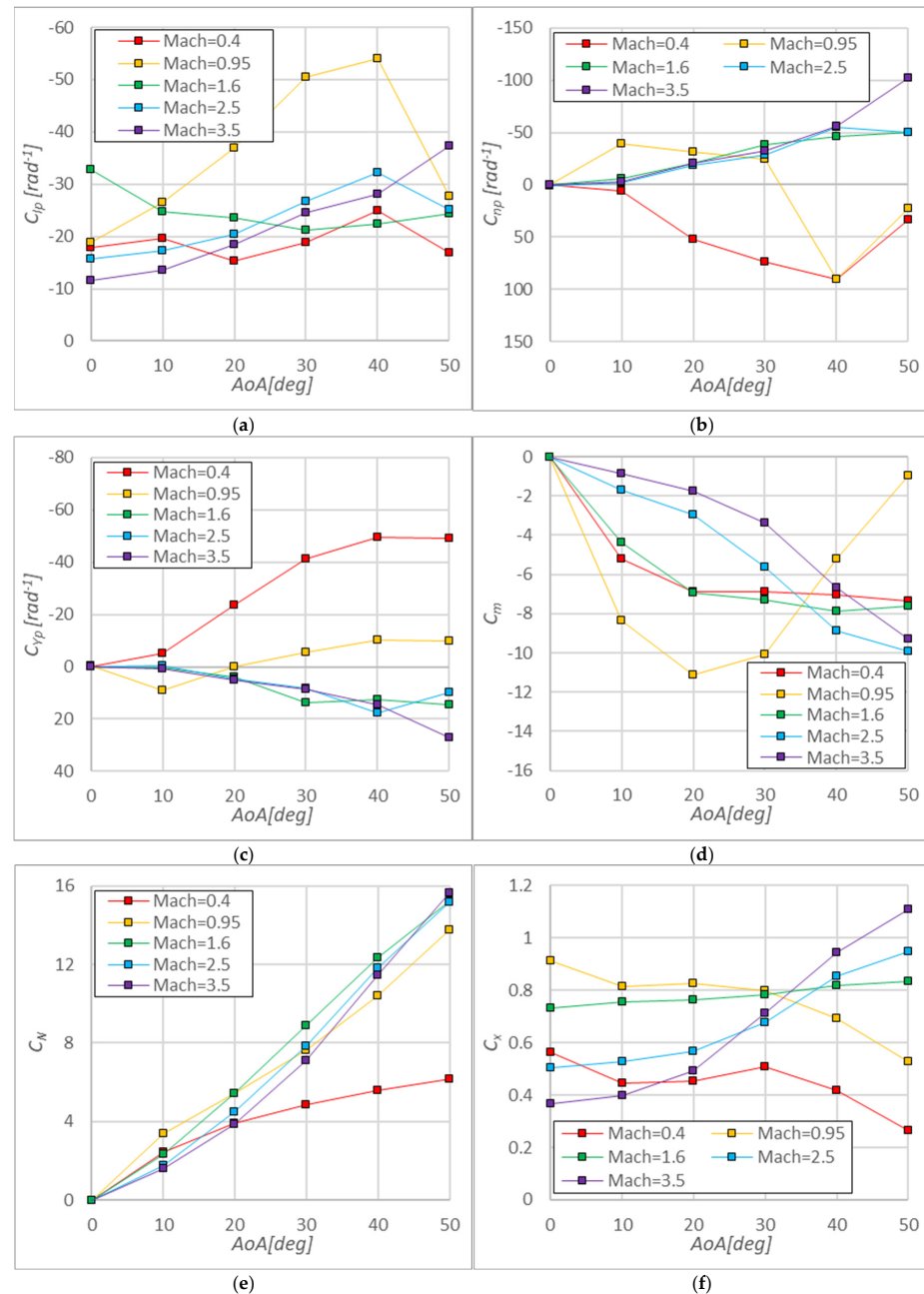


Figure 9. Aerodynamic coefficients variation with angle of attack at various Mach numbers: (a) C_{l_p} vs. AoA; (b) C_{n_p} vs. AoA; (c) C_{Y_p} vs. AoA; (d) C_m vs. AoA; (e) C_N vs. AoA; and (f) C_X vs. AoA; all data are averaged over roll angle.

The damping derivatives and pitch moment coefficient (C_{l_p} , C_{n_p} , C_{Y_p} , and C_m) did not present distinct variation trends, whereas the aerodynamic coefficients (C_N and C_X) followed a more consistent pattern.

The values of C_{l_p} remained negative, indicating stability on the roll axis, while C_{n_p} and C_{Y_p} presented both positive and negative values. The values of C_{l_p} varied from -54.03 rad^{-1} at Mach 0.95 and $\text{AoA} = 40^\circ$ to -11.61 rad^{-1} at Mach 3.5 and $\text{AoA} = 0^\circ$. In the subsonic case, C_{l_p} presented quasi-constant variation with AoA , remaining around -20 rad^{-1} , while in the transonic case, the variation with AoA presented a significant decrease from -18.87 rad^{-1} ($\text{AoA} = 0^\circ$) to -54.03 rad^{-1} ($\text{AoA} = 40^\circ$), followed by an increase to -27.66 rad^{-1} . The supersonic cases presented three distinct patterns: at Mach 1.6, C_{l_p} initially increased from -32.74 rad^{-1} ($\text{AoA} = 0^\circ$) to -21.14 rad^{-1} ($\text{AoA} = 30^\circ$), then decreased to -25.15 rad^{-1} ($\text{AoA} = 50^\circ$), while at Mach 2.5 it initially decreased from -15.71 rad^{-1} ($\text{AoA} = 0^\circ$) to -32.21 rad^{-1} ($\text{AoA} = 40^\circ$), followed by an increase to -25.15 rad^{-1} . For the Mach 3.5 regime, the trend strictly decreased from -11.61 rad^{-1} to 37.32 rad^{-1} .

The C_{n_p} remained positive for the subsonic case (Mach 0.4) and partially for the transonic case (Mach 0.95) at AoA above 30° , while negative values were observed for supersonic regimes. The supersonic cases presented a similar decreasing trend with comparable values up to 40° AoA . Beyond this point, Mach 1.6 and Mach 2.5 maintained constant values, whereas the values of C_{n_p} at Mach 3.5 continue to decrease.

In contrast, C_{Y_p} showed negative values for the subsonic case and partially for the transonic case at high incidences (above 20°), while positive values occurred in supersonic regimes and in the transonic regime at low incidences. Similar to C_{n_p} variation, C_{Y_p} variation in supersonic regimes showed a steadily increasing trend up to 40° AoA with similar values across Mach numbers. Beyond this point, only the Mach 3.5 regime continued to increase, while the Mach 1.6 and Mach 2.5 regimes remained constant.

The C_m was consistently negative (indicating nose-down behavior) across all Mach regimes, with negative slope ($\frac{\partial C_m}{\partial \alpha} < 0$), except for the transonic regime, where the slope became positive above 20° AoA .

The normal force coefficient C_N presented a consistent increasing trend across all Mach numbers. Meanwhile, the axial force coefficient C_X presented three distinct patterns: it decreased for subsonic and transonic cases, remained constant for Mach 1.6, and increased for Mach 2.5 and Mach 3.5.

3.3. Aerodynamic Coefficients Variation with Mach Number

Figure 10 presents the variation of C_{l_p} (Figure 10a), C_{n_p} (Figure 10b), C_{Y_p} (Figure 10c), C_m (Figure 10d), C_N (Figure 10e), and C_X (Figure 10f) as a function of angle of attack for different Mach numbers.

Similar to the previous case, the damping coefficients and pitch moment coefficient (C_{l_p} , C_{n_p} , C_{Y_p} , and C_m) did not present a distinct variation trend with Mach number, whereas the aerodynamic coefficients, C_N and C_X , showed a clear dependence on Mach variation. The values of C_{l_p} remained negative across all Mach numbers and AoA examined, while C_{n_p} values were predominantly negative in the supersonic regime and positive in subsonic and transonic regimes. In contrast to C_{n_p} , C_{Y_p} presented positive values in supersonic regimes and negative values in subsonic and transonic regimes. Additionally, C_m presented only negative values (nose down) at all incidences, attributed to the rear placement of the fins.

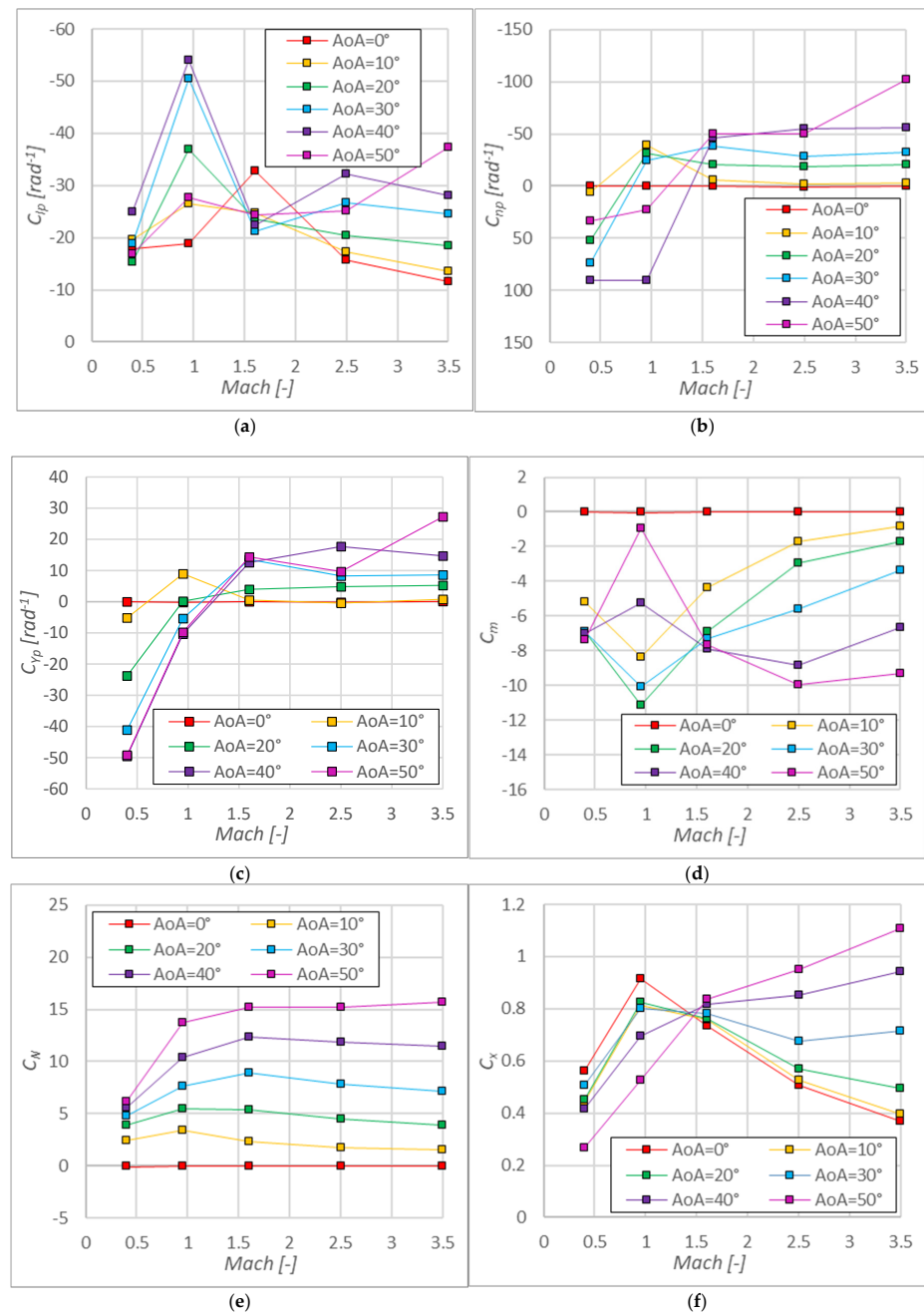


Figure 10. Aerodynamic coefficients variation with Mach number for various AoA: (a) C_{l_p} vs. Mach; (b) C_{n_p} vs. Mach; (c) C_{Y_p} vs. Mach; (d) C_m vs. Mach; (e) C_N vs. Mach; and (f) C_X vs. Mach; all data are averaged over roll angle.

The C_N presented a consistent increase from subsonic to transonic followed by a slight decrease and a stable pattern in supersonic regimes. C_X presented a significant increase from the subsonic to the transonic regime, followed by a slight decrease in the supersonic regime. However, at angles of attack of 40° and 50°, the axial force coefficient continued to increase even in supersonic regimes due to the formation of a bow shock on the ventral side of the model.

Figure 10 reveals that the values of C_{n_p} , C_{Y_p} , C_N , and C_m are zero at an angle of attack of 0°, which is a result of the model's double symmetry.

3.4. Aerodynamic Coefficients Variation with Roll Angle

3.4.1. Roll Moment Coefficient Variation

The variation in C_{l_p} with roll angle for various AoA is shown at Mach 0.4 (Figure 11a), at Mach 0.95 (Figure 11b), at Mach 1.6 (Figure 11c), at Mach 2.5 (Figure 11d), and at Mach 3.5 (Figure 11e).

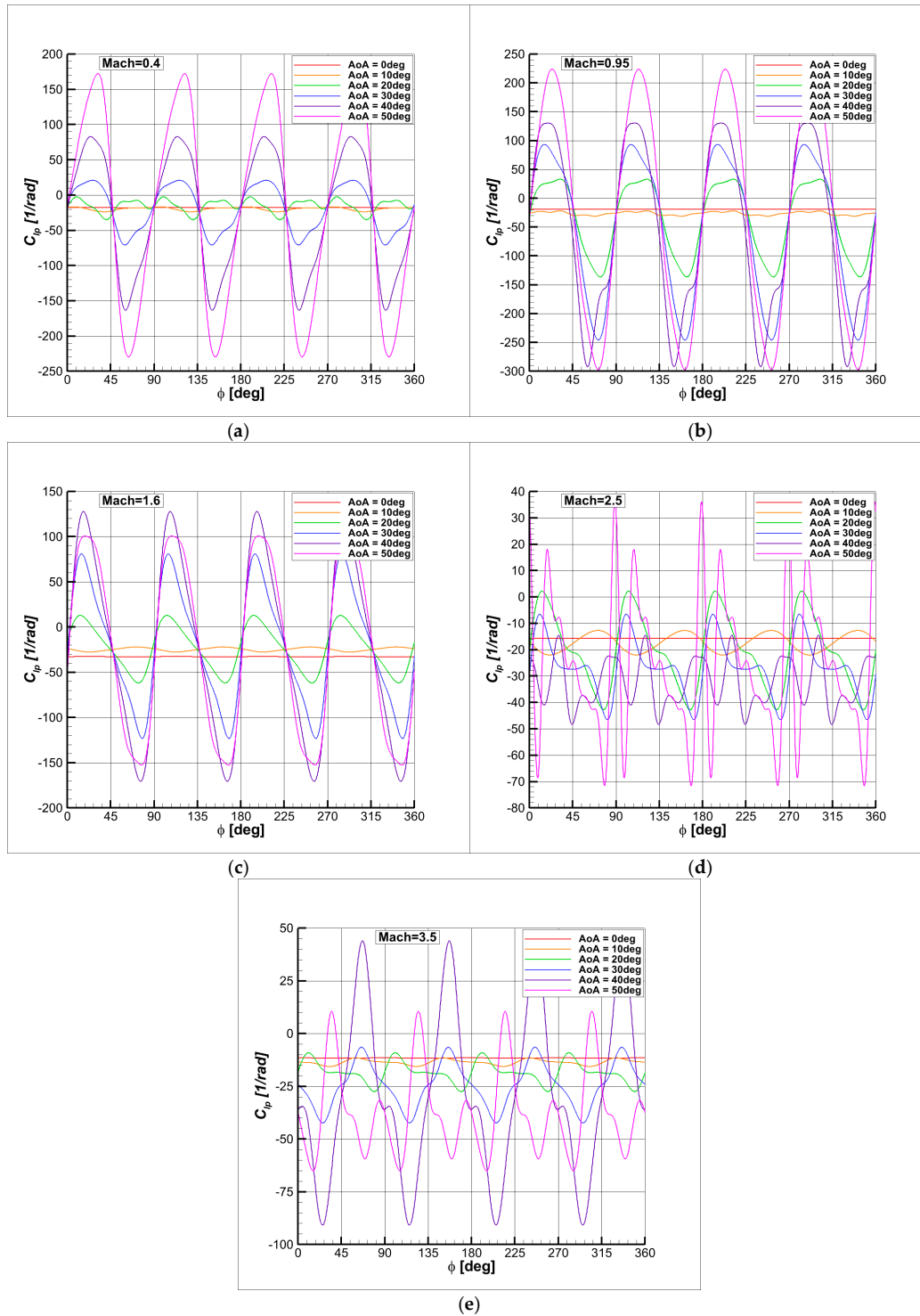


Figure 11. Roll moment coefficient variation with roll angle at various AoA: (a) C_{l_p} vs. ϕ at Mach 0.4; (b) C_{l_p} vs. ϕ at Mach 0.95; (c) C_{l_p} vs. ϕ at Mach 1.6; (d) C_{l_p} vs. ϕ at Mach 2.5; (e) C_{l_p} vs. ϕ at Mach 3.5.

Figure 11 shows that the roll damping moment coefficient had an oscillatory variation with respect to roll angle, with a periodicity of 90° due to the angular distance between the model's fins. For each case, the C_{l_p} maintained a constant, non-zero value at $\text{AoA} = 0^\circ$. Moreover, the oscillation amplitude of C_{l_p} increased with AoA . Despite the presence of positive values over certain roll angles, the mean value over a full roll cycle remained negative, indicating that the model maintained roll stability.

To better understand the nonlinear variation in C_{l_p} with incidence and Mach number, the variation with roll angle was also analyzed. While average C_{l_p} values were negative in all cases, the instantaneous C_{l_p} values were positive for some ranges of roll angles. The amplitudes of periodic C_{l_p} variations increased with incidence, except at 50° , where certain Mach numbers showed lower amplitudes compared to 40° . Amplitude variations with Mach number were roughly constant in subsonic and transonic regimes but decreased in supersonic regimes. Supersonic regimes presented additional oscillations compared to subsonic and transonic regimes due to shock wave interactions with the model.

The oscillation amplitudes of C_{l_p} varied across Mach numbers. In the subsonic case, the maximum amplitude occurred at 50° , ranging from -220 to 170 rad^{-1} . For the transonic case, the peak also appeared at 50° , with values ranging from -300 to 220 rad^{-1} . At Mach 1.6, the maximum amplitude occurred at 40° , ranging from -170 to 125 rad^{-1} , while at Mach 2.5, it peaked at 50° , with values between -70 and 35 rad^{-1} . Finally, for Mach 3.5, the maximum amplitude occurred at 40° , ranging from -90 to 45 rad^{-1} .

Another notable observation is that the C_{l_p} values for subsonic, transonic, and low supersonic conditions (Mach = 1.6) tended to converge to similar values, regardless of the angle of attack, when the roll angle of the model was 45° ($\times$ position) and 90° ($+$ position). Moreover, the oscillation curves at Mach 0.4 for $\text{AoA} = 10^\circ$ and 20° , at Mach 1.6 for $\text{AoA} = 10^\circ$, and at Mach 2.5 for $\text{AoA} = 10^\circ$ were out of phase with respect to the curves at other angles of attack. These specific curves exhibited values above the average when others were below, and vice versa, indicating a phase shift in their oscillatory behavior. The transonic regime presented consistent and coherent variation, where all curves tended to rise above or fall below the average simultaneously. In contrast, the high supersonic regime displayed highly nonlinear behavior, making it difficult to identify a clear or consistent pattern.

The flow conditions below Mach 1.6 presented a consistent pattern, with the oscillation curves showing a predominantly sinusoidal variation. In contrast, the higher supersonic regimes, above Mach 2.5, displayed irregular and highly oscillatory behavior that lacked coherence across different angles of incidence.

3.4.2. Yaw Moment Coefficient Variation

The variation in C_{n_p} with roll angle for various AoA is shown at Mach 0.4 (Figure 12a), at Mach 0.95 (Figure 12b), at Mach 1.6 (Figure 12c), at Mach 2.5 (Figure 12d), and at Mach 3.5 (Figure 12e).

The variation in C_{n_p} with roll angle followed a similar variation trend to that of C_{l_p} . At $\text{AoA} = 0^\circ$, C_{n_p} remained constant null, while at non-zero AoA , C_{n_p} presented an oscillatory pattern with both positive and negative values. This behavior was due to the variations in effective incidence, as well as interactions between the fins and shock waves, detached flow, and vortex structures. The oscillation period remained 90° , corresponding to the angle between the model's fins. The average values were not only negative in this case, indicating that the model maintained yaw stability only for specific Mach numbers and angles of attack, as presented in Figures 9b and 10b.

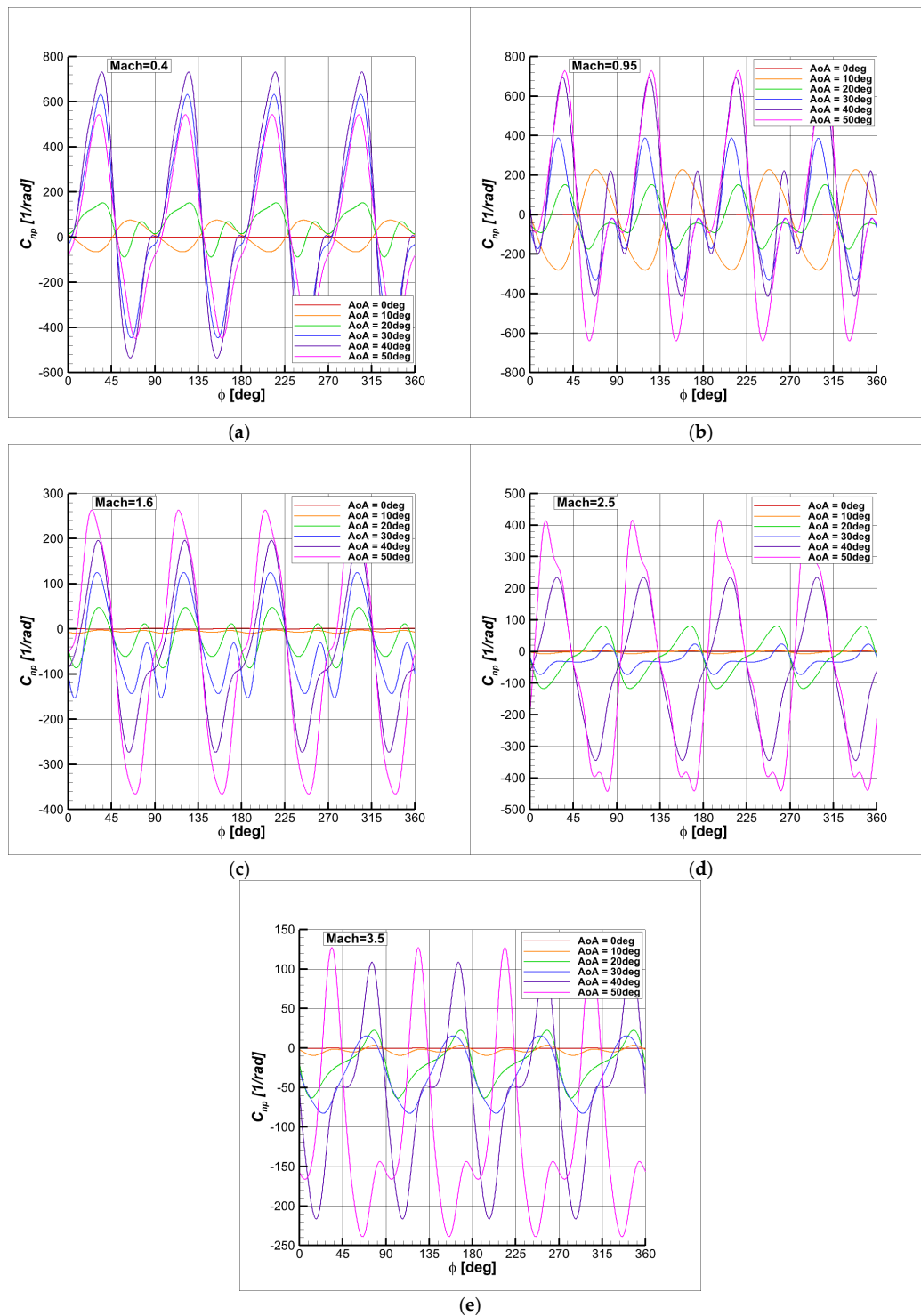


Figure 12. Yaw moment coefficient variation with roll angle at various AoA: (a) C_{n_p} vs. ϕ at Mach 0.4; (b) C_{n_p} vs. ϕ at Mach 0.95; (c) C_{n_p} vs. ϕ at Mach 1.6; (d) C_{n_p} vs. ϕ at Mach 2.5; (e) C_{n_p} vs. ϕ at Mach 3.5.

The subsonic case shows that, as the angle of attack (AoA) increased, the oscillation amplitudes also increase up to $\text{AoA} = 40^\circ$, while the mean value gradually increased. This behavior indicates a stronger unsteady aerodynamic response at higher angles of attack in subsonic conditions.

In the transonic regime, the mean value presented an alternating trend, shifting between positive and negative values, suggesting the presence of complex flow phenomena

such as shock wave interactions and flow separation. Additionally, the amplitude of oscillations increases with AoA, reflecting enhanced unsteady aerodynamic effects.

For supersonic cases, the mean values decreased continuously with incidence, indicating a stronger unsteady aerodynamic response. Moreover, the oscillation amplitudes continued to increase with AoA, suggesting that the unsteady fluctuations become more pronounced at higher angles of attack.

Figure 12 illustrates that the range of variation differed across Mach numbers and angles of attack. For instance, focusing on the largest range in each case, Mach 0.4 presented its widest variation at AoA = 40°, with limits from −520 to 750 rad^{−1}. In contrast, the other regimes reach their maximum variation at AoA = 50°, with the following ranges: Mach 0.95 from −640 to 720 rad^{−1}, Mach 1.6 from −360 to 260 rad^{−1}, Mach 2.5 from −440 to 400 rad^{−1}, and Mach 3.5 from −240 to 140 rad^{−1}.

In contrast to C_{l_p} , the variation in C_{n_p} displayed a more consistent and predominantly sinusoidal trend. In some cases, the curves appeared to be a combination of multiple sinusoidal components. Nevertheless, particularly in the higher supersonic regimes, the variation patterns remained clear and well-ordered. Moreover, in all cases except Mach 3.5, the curves intersected consistently at roll angles of 45° (× position) and 90° (+ position).

The oscillation phase also varied across several flight conditions. Specifically, for Mach 0.4 at AoA = 10°, Mach 0.95 at AoA = 10°, Mach 2.5 at AoA = 40–50°, and Mach 3.5 at AoA = 50°, the curves exhibited values below the mean when others were above, and vice versa. This behavior suggests a phase shift of approximately 45°.

3.4.3. Side Force Coefficient Variation

The variation in C_{Y_p} with roll angle for various AoA is presented at Mach 0.4 (Figure 13a), at Mach 0.95 (Figure 13b), at Mach 1.6 (Figure 13c), at Mach 2.5 (Figure 13d), and at Mach 3.5 (Figure 13e).

Similarly to the previous case (C_{n_p} vs. ϕ), the C_{Y_p} remained zero at AoA = 0° because the model's fins experienced the same effective incidences, resulting in no net side force. However, as the AoA deviated from 0°, the C_{Y_p} began to vary, presenting an oscillatory pattern with alternating positive and negative values and with a 90° period. In addition, increasing the incidence resulted in an increase in the amplitude of oscillation. Additionally, significant variations in C_{Y_p} were observed at certain roll angles due to complex interactions between the fins and flow phenomena such as shock waves, detached flow, and vortex structures.

In the subsonic case, C_{Y_p} presented an oscillatory variation, with amplitudes increasing as the angle of attack (AoA) rose to 40°, after which the amplitude began to decrease. At angles of attack of 30°, 40°, and 50°, the oscillation patterns were similar, presenting close amplitude values. However, at 10° and 20° AoA, a phase shift of approximately 45° was observed, attributed to the onset of flow detachment occurring beyond 10° AoA.

The transonic case followed a similar oscillatory trend as the subsonic regime, both in terms of oscillation patterns and amplitude variations. The mean values of oscillations tend to remain negative for a significant range of incidences, similar to the subsonic case.

For supersonic cases, larger oscillation amplitudes were observed at AoA values of 40° and 50°, whereas at 10° AoA, the amplitude remained very small. Additionally, it was noted that the mean values of oscillations tended to increase with AoA, indicating a shift in aerodynamic response as the angle of attack rose. Also, the mean values were strictly positive, as presented in Figure 9.

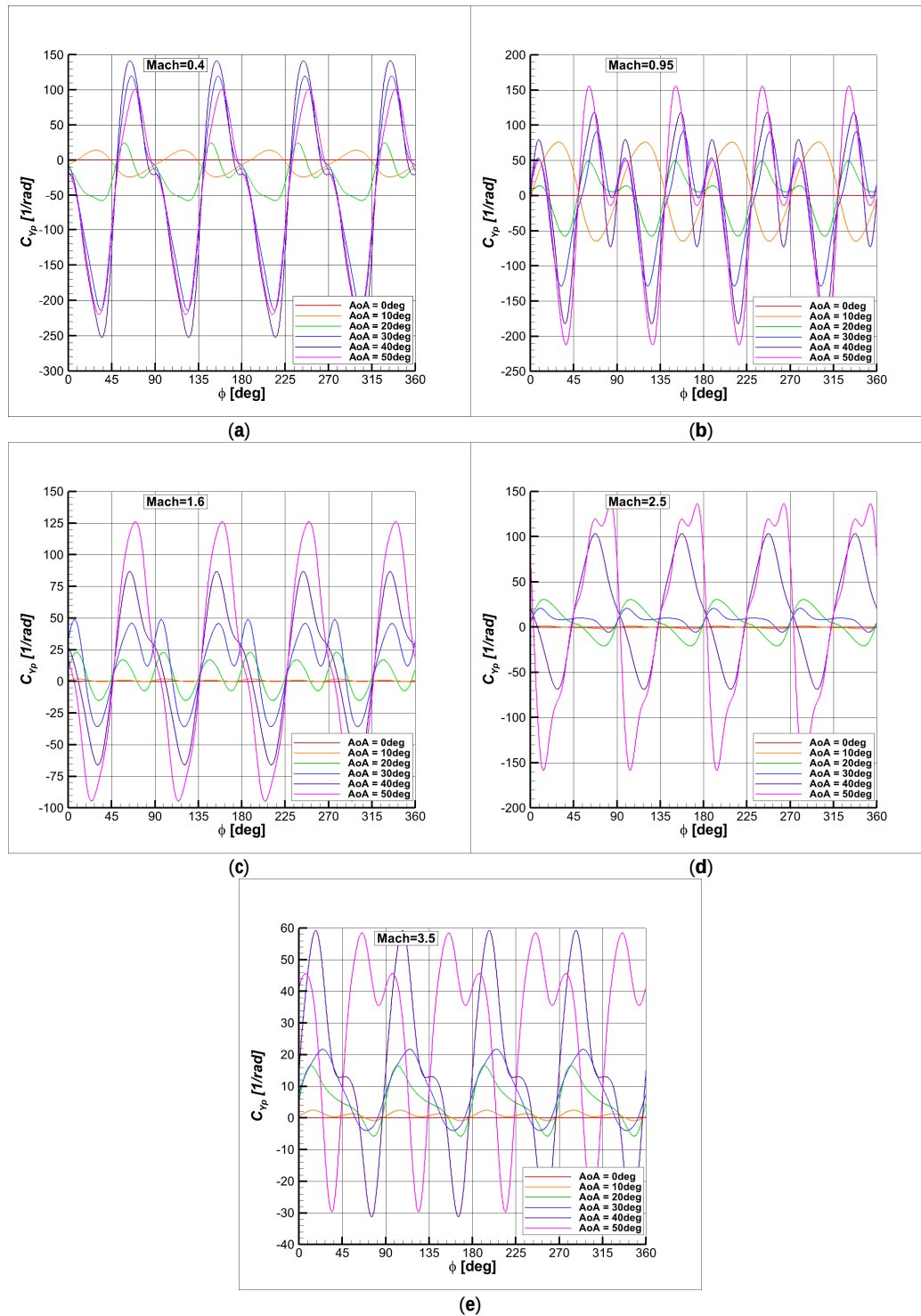


Figure 13. Side force coefficient variation with roll angle at various AoA: (a) C_{Y_p} vs. ϕ at Mach 0.4; (b) C_{Y_p} vs. ϕ at Mach 0.95; (c) C_{Y_p} vs. ϕ at Mach 1.6; (d) C_{Y_p} vs. ϕ at Mach 2.5; (e) C_{Y_p} vs. ϕ at Mach 3.5.

Figure 13 illustrates that the range of variation differed across Mach numbers and angles of attack. For instance, focusing on the largest range in each case, Mach 0.4 presented its widest variation at $\text{AoA} = 40^\circ$, with limits from -260 rad^{-1} to 140 rad^{-1} . Similarly, Mach = 3.5 presented its maximal range at $\text{AoA} = 40^\circ$, with limits between -30 rad^{-1} and 60 rad^{-1} . In contrast, the other regimes reach their maximum variation at $\text{AoA} = 50^\circ$, with the following ranges: Mach 0.95 from -220 rad^{-1} to 160 rad^{-1} , Mach 1.6 from -95 rad^{-1} and 125 rad^{-1} , and Mach 2.5 from -160 rad^{-1} and 140 rad^{-1} .

3.4.4. Pitch Moment Coefficient Variation

The variation in C_m with roll angle for various AoA is shown at Mach 0.4 (Figure 14a), Mach 0.95 (Figure 14b), Mach 1.6 (Figure 14c), Mach 2.5 (Figure 14d), and Mach 3.5 (Figure 14e).

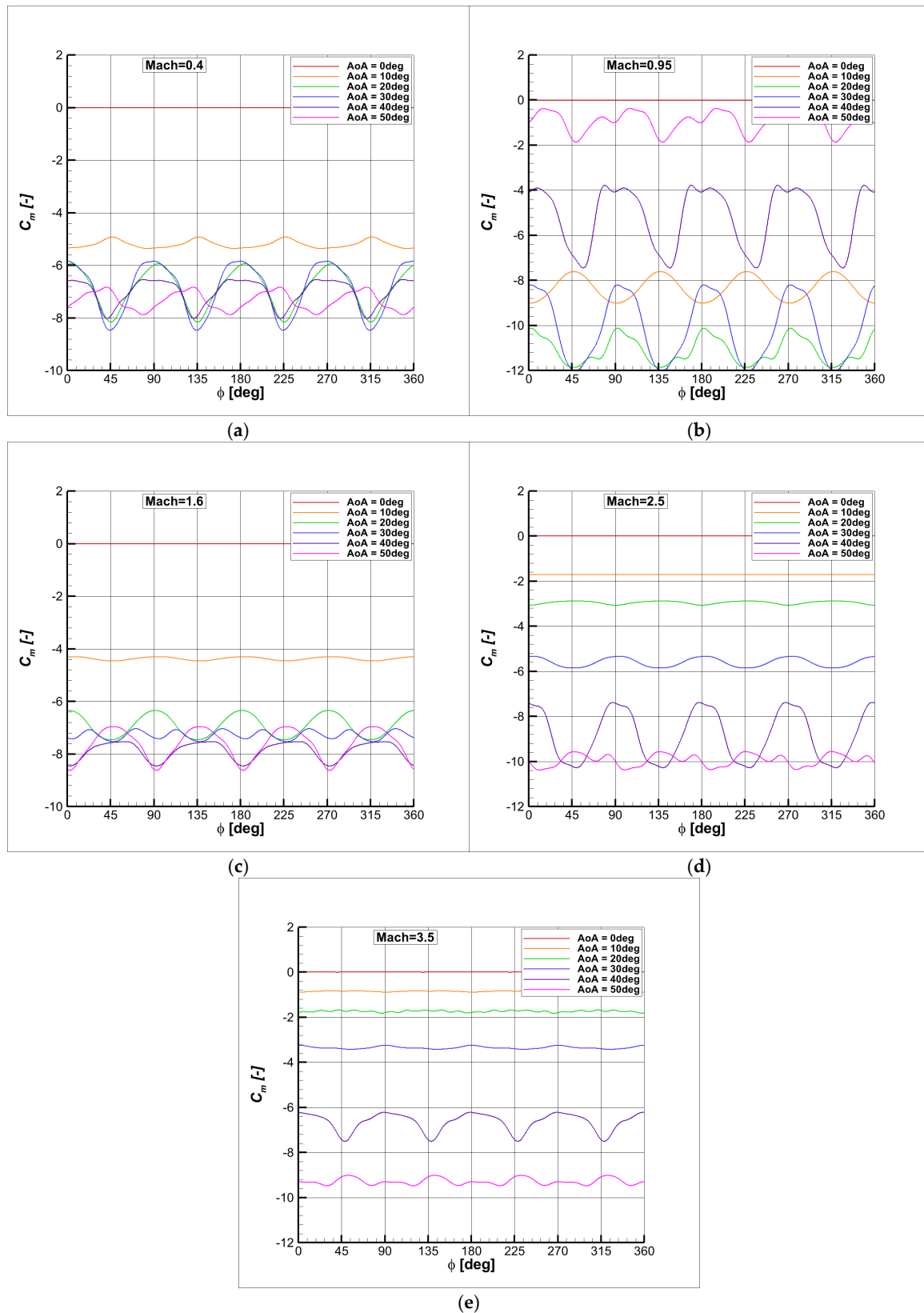


Figure 14. Side force coefficient variation with roll angle at various AoA: (a) C_m vs. ϕ at Mach 0.4; (b) C_m vs. ϕ at Mach 0.95; (c) C_m vs. ϕ at Mach 1.6; (d) C_m vs. ϕ at Mach 2.5; (e) C_m vs. ϕ at Mach 3.5.

Figure 14 shows that C_m presented an approximately sinusoidal variation, except at high angles of attack, where the oscillations appeared to result from a superposition of sinusoidal components. This behavior is likely attributable to complex interactions between the fins and flow phenomena such as shock waves, flow separation, and vortex formation.

At low incidence angles or for the high supersonic regimes, the average values of C_m tended to decrease as the angle of incidence increased. In contrast, within the subsonic and low supersonic regimes, the mean values of C_m remained approximately constant for incidence angles above 20° , and in the transonic regime, C_m began to increase with incidence beyond this point.

Figure 14 indicates that C_m values in the subsonic and low supersonic regimes were closely aligned, as were the C_m values in the high supersonic regimes, as shown in Figure 9d.

Across all flow regimes, the moment coefficient (C_m) remained consistently negative, as expected, given that the only lifting surfaces were the fins positioned at the rear of the model. This configuration naturally induces a nose-down pitching moment.

Additionally, it was observed that the mean values C_m decreased in magnitude with the angle of attack (AoA), along with an increase in the amplitude of oscillations, indicating a growing unsteady aerodynamic influence at higher AoA values.

The minimum values of C_m occurred at an incidence angle of 30° in the subsonic and transonic regimes, whereas in the supersonic regimes, the minimum values were observed at an incidence angle of 50° . The maximum oscillation amplitudes of C_m were observed as follows: for Mach 0.4, C_m ranged from -5.9 to -8.5 ; for Mach 0.95, from -8.2 to -12.1 ; for Mach 1.6, from -6.9 to -8.5 ; for Mach 2.5, from -7.5 to -10.3 ; and for Mach 3.5, from -6.2 to -7.5 .

3.4.5. Normal Force Coefficient Variation

The variation of normal force C_N with roll angle for various AoA is shown at Mach 0.4 (Figure 15a), Mach 0.95 (Figure 15b), Mach 1.6 (Figure 15c), Mach 2.5 (Figure 15d), and Mach 3.5 (Figure 15e).

The normal force coefficient presented a quasi-constant variation trend, especially at angles of attack of up to 30° . At higher angles of attack, the interaction between the fins and shock waves, detached flow, and vortex structures introduced small oscillations with a 90° periodicity in roll angle. However, these oscillations presented a sinusoidal pattern and did not overlap across the range of angles of attack. Moreover, the amplitude of oscillations increased with angle of attack.

At AoA = 0° , the values of normal force coefficient were zero and increased constantly with angle of attack, as presented in Figure 9e. The values in the transonic and supersonic regimes remained closely aligned, while the subsonic regime showed significantly lower values of normal force coefficient. Additionally, C_N values in the transonic and supersonic regimes were quite similar, whereas in the subsonic regime, they were notably lower.

In several cases, it can be observed that the phase of oscillations differed from the others. At Mach 0.4 (AoA = 10° and AoA = 50°), Mach 0.95 (AoA = 10°), Mach 1.6 (AoA = 40° and AoA = 50°), and Mach 2.5 (AoA = 50°), the values of C_N were above average, while in the other cases, they are below average, and vice versa.

The maximum oscillation ranges were observed at different incidence angles across various Mach numbers. At Mach 0.4, the maximum range was between 4.4 and 5.3 at an incidence of 30° . At Mach 0.95, the maximum range spanned from 9.9 to 10.6 at 40° incidence. For Mach 1.6, the maximum range was between 5.2 and 5.6 at 20° incidence. At Mach 2.5, the maximum range fell between 11.4 and 12.2 at 40° incidence, while at Mach 3.5, the maximum range was between 15.2 and 16.1 at 50° incidence.

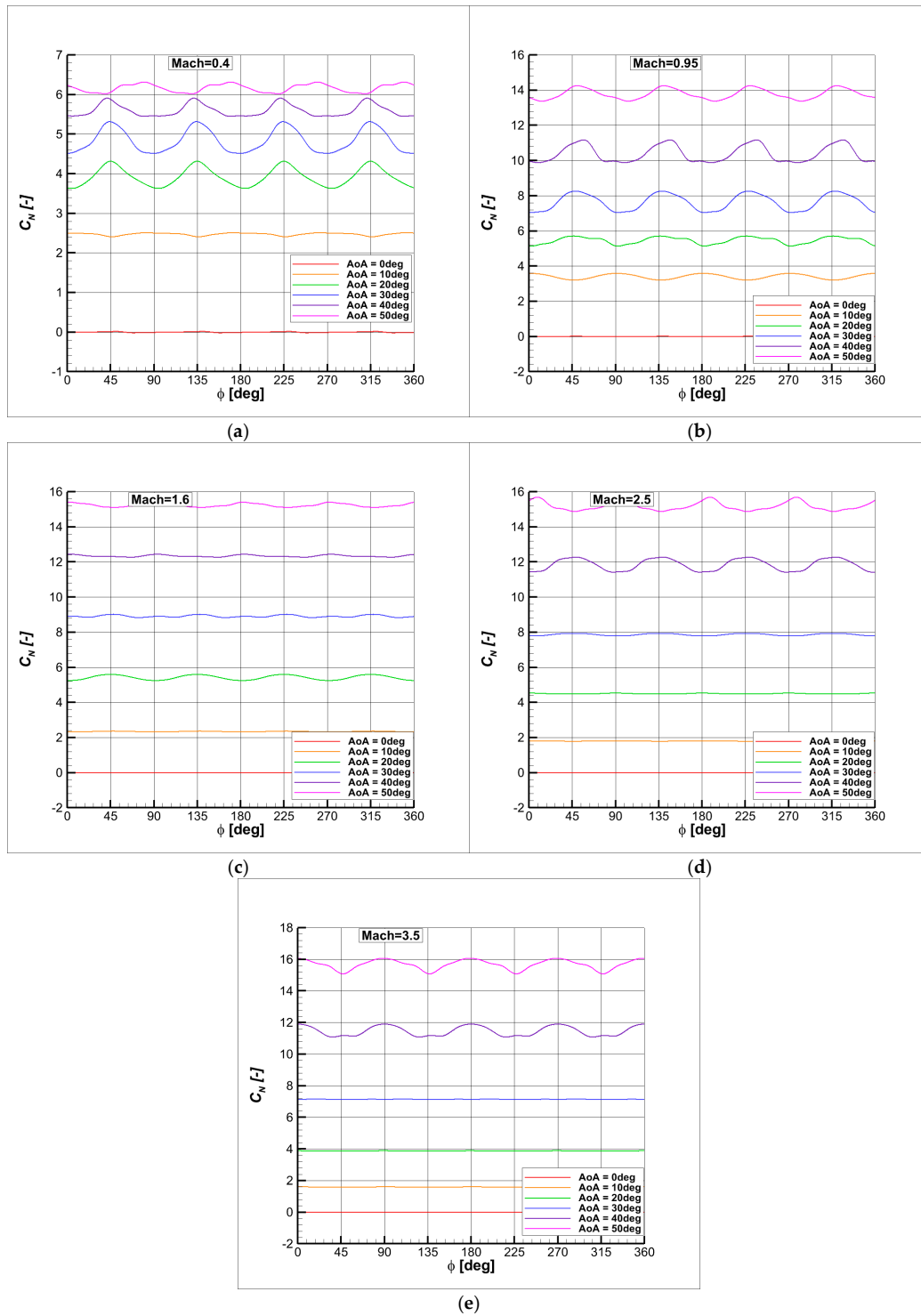


Figure 15. Normal force coefficient variation with roll angle at various AoA: (a) C_N vs. ϕ at Mach 0.4; (b) C_N vs. ϕ at Mach 0.95; (c) C_N vs. ϕ at Mach 1.6; (d) C_N vs. ϕ at Mach 2.5; and (e) C_N vs. ϕ at Mach 3.5.

3.4.6. Axial Force Coefficient Variation

The variation of C_X with roll angle for various AoA is shown at Mach 0.4 (Figure 16a), at Mach 0.95 (Figure 16b), at Mach 1.6 (Figure 16c), at Mach 2.5 (Figure 16d), and at Mach 3.5 (Figure 16e).

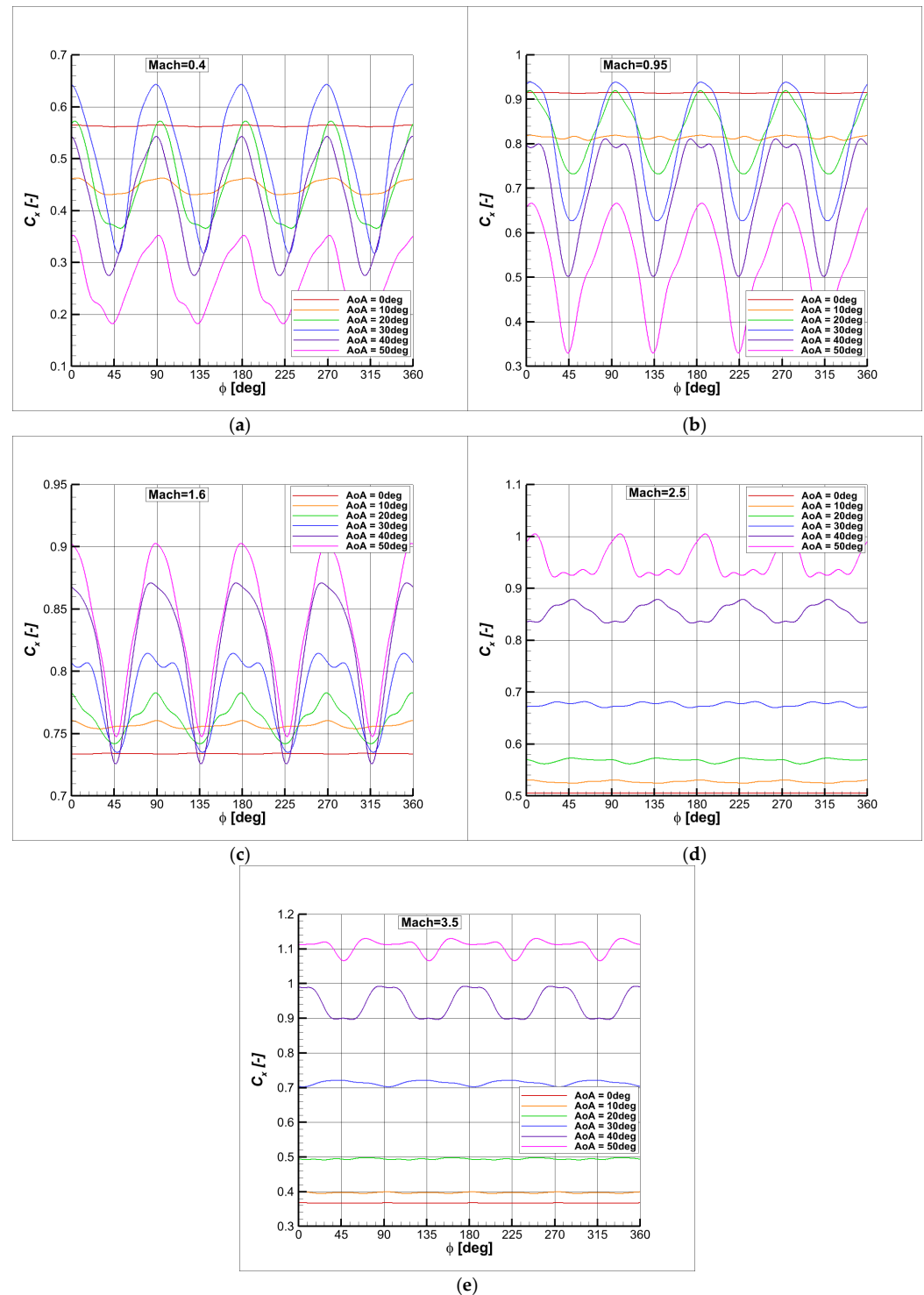


Figure 16. Axial force coefficient variation with roll angle at various AoA: (a) C_X vs. ϕ at Mach 0.4; (b) C_X vs. ϕ at Mach 0.95; (c) C_X vs. ϕ at Mach 1.6; (d) C_X vs. ϕ at Mach 2.5; (e) C_X vs. ϕ at Mach 3.5.

The axial force coefficient presented an oscillatory variation trend, especially in the subsonic, transonic, and low-supersonic regimes. In the high supersonic regimes, the oscillation amplitudes were smaller, while the average values increased steadily with increasing incidence. In subsonic and transonic regimes, the axial force coefficient decreased when the AoA increased, but for the supersonic regimes the values of C_X increased as the AoA increased.

In the subsonic regime, the axial force coefficient (C_X) remained constant at 0° AoA, while its mean values decreased as the angle of attack increased, accompanied by growing

oscillation amplitudes. A similar trend was observed in the transonic regime, with the key distinction that C_X values were higher due to the influence of shock waves.

In the supersonic regime, the mean values of C_X increased with AoA, following the same trend as the oscillation amplitudes, indicating a stronger aerodynamic response at higher angles of attack.

The maximum oscillation ranges occurred at different incidence angles for each Mach number. At Mach 0.4, the range extended from 0.32 to 0.64 at an incidence angle of 30°. For Mach 0.95, the maximum range was from 0.32 to 0.67 at 50°. At Mach 1.6, the range lay between 0.75 and 0.90, also at 50° incidence. In the case of Mach 2.5, the range spanned from 0.92 to 1.01 at 50°, while at Mach 3.5, the maximum range was observed between 0.90 and 0.98 at an incidence angle of 40°.

3.5. Mach Number and Pressure Coefficient Distribution

3.5.1. Mach 0.4

The following figures present the distribution of pressure coefficient (C_p) on the model surface and the distribution of Mach number on the vertical plane (XoZ) at 0° roll angle, Mach 0.4, and the following angles of attack: 0° (Figure 17a), 10° (Figure 17b), 20° (Figure 17c), 30° (Figure 17d), 40° (Figure 17e), and 50° (Figure 17f).

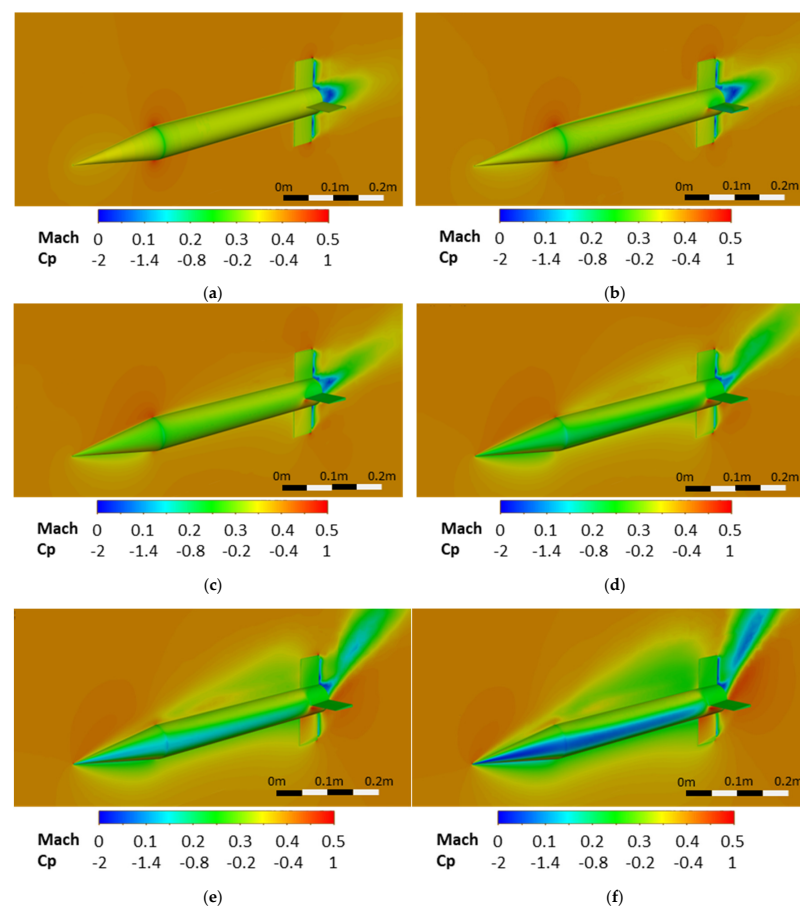


Figure 17. Mach number and pressure coefficient distribution at Mach 0.4 and various AoA: (a) AoA = 0°; (b) AoA = 10°; (c) AoA = 20°; (d) AoA = 30°; (e) AoA = 40°; (f) AoA = 50°.

The pressure coefficient C_p is defined as $C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V^2}$ and represents a dimensionless number that describes the relative pressure on a surface point compared to the free-stream dynamic pressure.

At Mach 0.4, the flow around the model remained subsonic, with Mach number values between 0 and 0.5, while the pressure coefficient varied between -2 and 1 .

The wake generated behind the model was minimal at 0° , and it is observed that the suction zone expanded as the AoA increased, leading to an increase in drag coefficient. Additionally, at higher angles of attack, detached flow on the dorsal surface of the model was observed, which reduced the effectiveness of the dorsal fin. At these higher angles of attack, the stagnation area increased, forcing the fluid to accelerate on the side of the model.

Moreover, at high angles of attack (AoA), a lateral acceleration of the flow around the model was observed, leading to the formation of vortices that interacted with the rear-mounted fins. Additionally, due to the incidence angles, the lower surfaces of the fins exhibited a high pressure coefficient (C_p), which counteracted the rolling motion when the fin moved downward and enhanced it when the fin moved upward. Also, this increase of pressure coefficient on the lower surface of the lateral fins generated a nose-down pitch moment.

3.5.2. Mach 0.95

The figures below present the distribution of pressure coefficient (C_p) on the model surface and the distribution of Mach number on the vertical plane (XoZ) at a 0° roll angle, Mach 0.95, and the following angles of attack: 0° (Figure 18a), 10° (Figure 18b), 20° (Figure 18c), 30° (Figure 18d), 40° (Figure 18e), and 50° (Figure 18f).

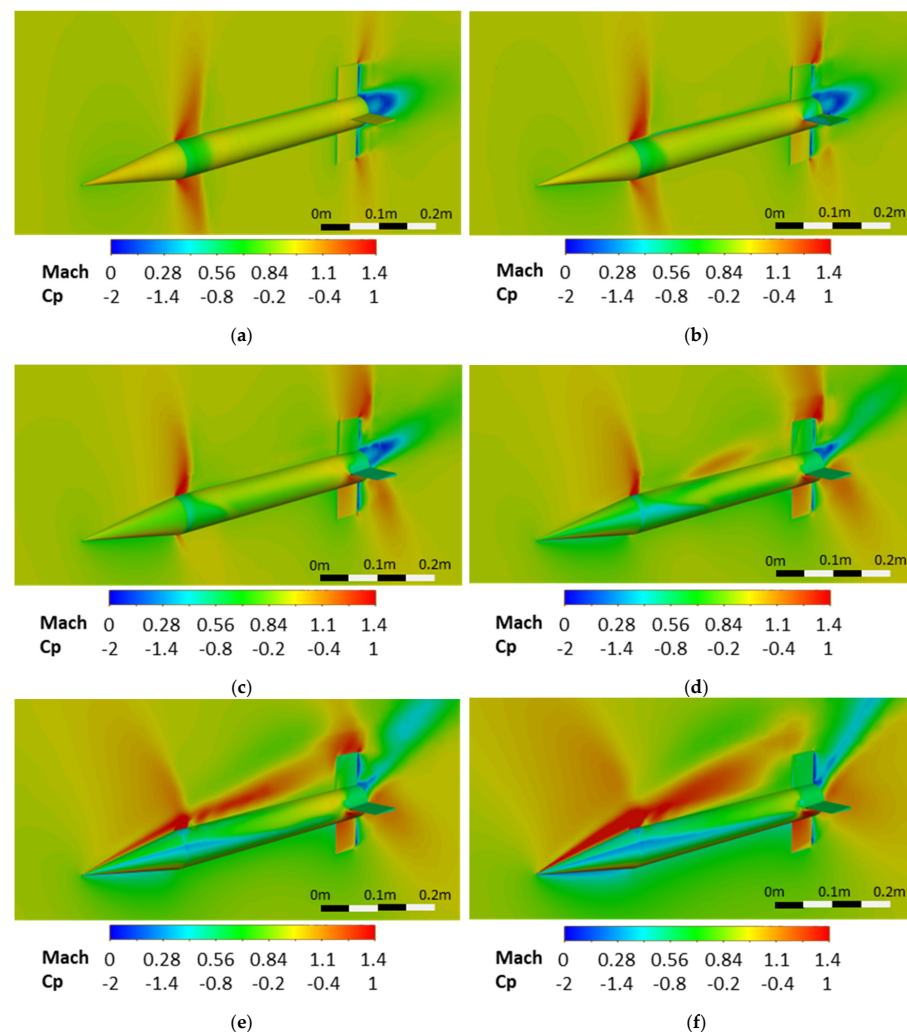


Figure 18. Mach number and pressure coefficient distribution at Mach 0.95 and various AoA: (a) AoA = 0° ; (b) AoA = 10° ; (c) AoA = 20° ; (d) AoA = 30° ; (e) AoA = 40° ; and (f) AoA = 50° .

At Mach 0.95, the flow around the model presented Mach numbers between 0 and 1.4, while the pressure coefficient varied between -2 and 1. The cone section of the model compressed the flow, while the cylindrical fuselage section generated an expansion fan accelerating the flow until the shock wave formed.

As the angle of attack increased, the shock wave shifted forward and became more pronounced. Additionally, at higher angles of attack, detached flow was observed on the dorsal surface of the model, diminishing the effectiveness of the dorsal fin. Furthermore, at these higher angles, the stagnation area increased, causing the fluid to accelerate along the sides of the model. The lateral acceleration of the flow induced separation vortices that interacted with the fins at certain roll angles. Additionally, this lateral acceleration led to the formation of an intense supersonic flow along the plane of symmetry.

3.5.3. Mach 1.6

The figures below show the distribution of pressure coefficient (C_p) on the model surface and the distribution of Mach number on the vertical plane (XoZ) at a 0° roll angle, Mach 1.6, and the following angles of attack: 0° (Figure 19a), 10° (Figure 19b), 20° (Figure 19c), 30° (Figure 19d), 40° (Figure 19e), and 50° (Figure 19f).

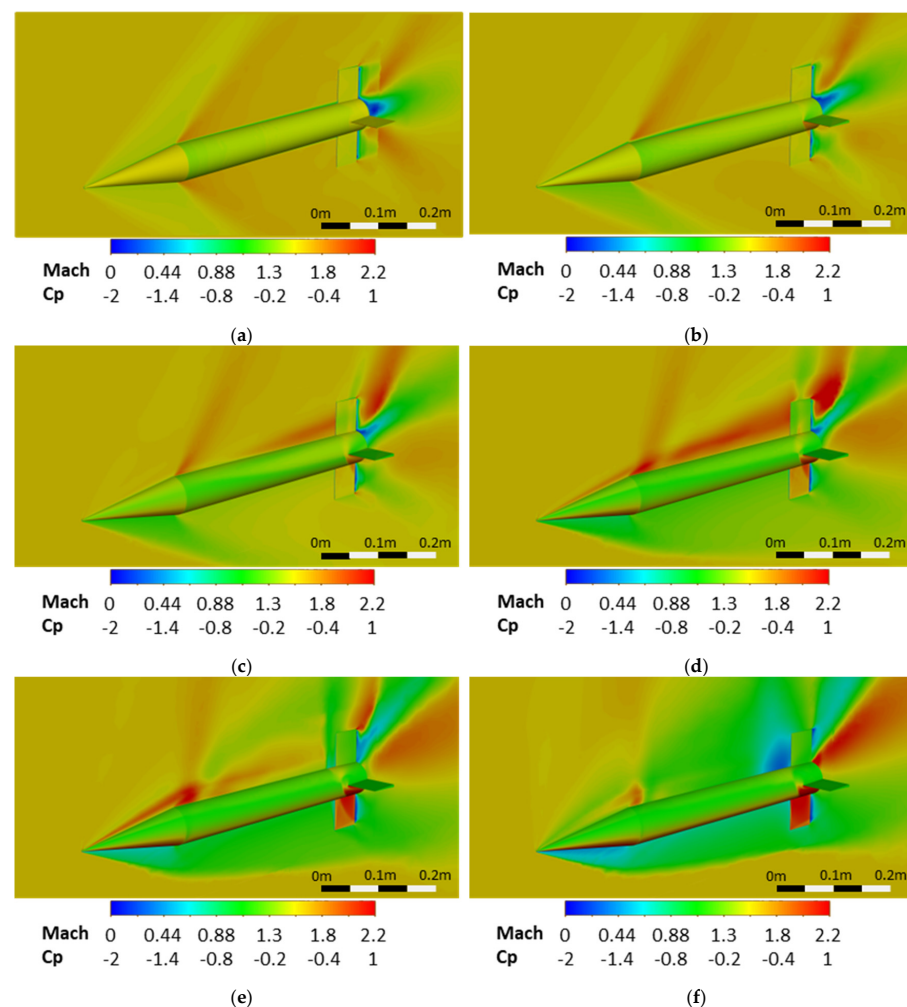


Figure 19. Mach number and pressure coefficient distribution at Mach 1.6 and various AoA: (a) AoA = 0° ; (b) AoA = 10° ; (c) AoA = 20° ; (d) AoA = 30° ; (e) AoA = 40° ; and (f) AoA = 50° .

At Mach 1.6, the flow around the model displayed Mach numbers ranging from 0 to 2.2, while the pressure coefficient varied between -2 and 1. At an incidence of 0° , the shock wave originated at the tip of the model, with the fluid being decelerated up to the transition

region between the cone and the cylinder, where it accelerated due to the expansion fan. As the incidence increased, the fluid decelerated on the ventral side of the model. Additionally, up to an incidence of 40° , the dorsal side of the model experienced acceleration due to both axial and lateral flow.

The ventral fin and the ventral faces of the lateral fins presented a higher-pressure coefficient, which caused oscillatory variation in the aerodynamic coefficients with the roll angle. Furthermore, from an incidence of 10° onward, the shock wave generated by the lateral fin interacted with the ventral fin, resulting in an increase in pressure.

At Mach 1.6, the ventral bow shock was positioned far enough from the body to prevent interaction between the fins and the undisturbed incoming flow. As a result, the ventral fins were located behind the shock, where the airflow was subsonic, and thus experienced a lower airspeed. In contrast, the dorsal fins remained exposed to supersonic flow and generated their own shock waves. This difference explains why the variations in damping derivatives at Mach 1.6 more closely resemble those seen in subsonic and transonic regimes, rather than in higher supersonic conditions, as shown in Figures 11–13.

3.5.4. Mach 2.5

The following figures illustrate the distribution of pressure coefficient (C_p) on the model surface and the distribution of Mach number on the vertical plane (XoZ) at a 0° roll angle, Mach 2.5, and the following angles of attack: 0° (Figure 20a), 10° (Figure 20b), 20° (Figure 20c), 30° (Figure 20d), 40° (Figure 20e), and 50° (Figure 20f).

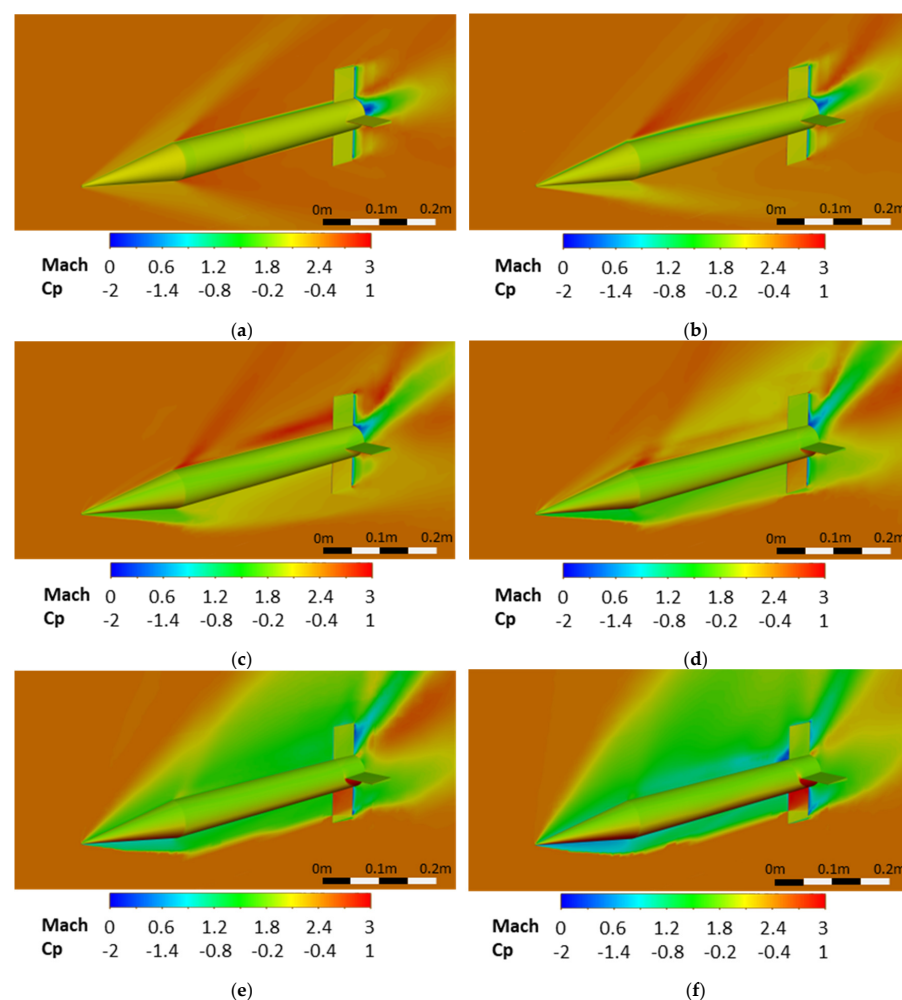


Figure 20. Mach number and pressure coefficient distribution at Mach 2.5 and various AoA: (a) AoA = 0° ; (b) AoA = 10° ; (c) AoA = 20° ; (d) AoA = 30° ; (e) AoA = 40° ; and (f) AoA = 50° .

At Mach 2.5, the flow around the model showed Mach numbers ranging from 0 to 3, while the pressure coefficient varied between -2 and 1 . As the Mach number increased, the angle between the shock wave and the surface of the model decreased. Furthermore, as the incidence increased, the shock wave intensified due to growth of the ventral stagnation area. Fluid acceleration on the dorsal side of the model was present only up to an incidence of 30° , where it remained relatively weak. Beyond an incidence of 30° , the flow in the dorsal region became subsonic and detached.

Furthermore, the model's fins interacted with the shock wave generated by the body of the model, causing pressure variation in the ventral region, after an incidence of 40° . This interaction generated a step change in the variations in the aerodynamic coefficients with roll angle. Additionally, the dorsal fin and the dorsal surfaces of the lateral fins experienced a decrease in pressure as the incidence increased.

The partial exposure of the ventral fins to undisturbed flow at high angles of attack led to an unusual variation in the damping derivatives, distinct from those observed at lower Mach numbers. The exposed fin surfaces generated shock waves that interacted and disrupted the sinusoidal nature of the roll variation.

3.5.5. Mach 3.5

The figures below illustrate the distribution of pressure coefficient (C_p) on the model surface and the distribution of Mach number on the vertical plane (XoZ) at a 0° roll angle, Mach 3.5, and the following angles of attack: 0° (Figure 21a), 10° (Figure 21b), 20° (Figure 21c), 30° (Figure 21d), 40° (Figure 21e), and 50° (Figure 21f).

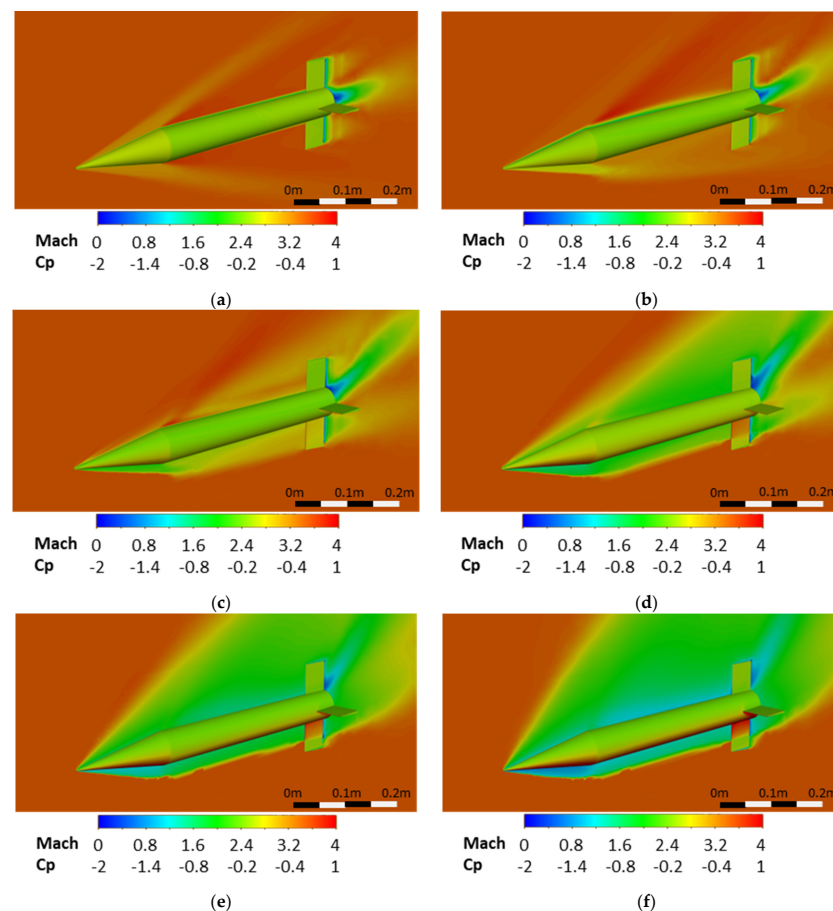


Figure 21. Mach number and pressure coefficient distribution at Mach 3.5 and various AoA: (a) AoA = 0° ; (b) AoA = 10° ; (c) AoA = 20° ; (d) AoA = 30° ; (e) AoA = 40° ; and (f) AoA = 50° .

At Mach 3.5, the flow around the model presented Mach numbers ranging from 0 to 4, while the pressure coefficient varied between -2 and 1 . As the incidence increased, the intensity of the shock wave on the ventral side of the model grew, while on the dorsal side, the shock wave intensity decreased until an incidence of 20° , when the shock wave began to intensify due to the flow separation from the body of the model.

Furthermore, it was observed that the dorsal fin did not present pressure coefficient variations at different incidences, while the ventral fin showed an increase in the pressure coefficient with incidence up to 20° . Beyond this point, a strong interaction with the ventral shock wave generated by the body of the model was observed. This interaction induced a sudden pressure variation, transitioning from high pressure near the base to low pressure near the tip. These sudden changes generated abrupt variations in the aerodynamic coefficients with the roll angle.

A similar phenomenon to that observed at Mach 2 also occurred at Mach 3.5 on the ventral side at high angles of attack. The partial exposure of the model's fins to the undisturbed flow generated shock waves that disrupted the sinusoidal variation of damping derivatives with roll angle. At this higher Mach number, the ventral bow shock was tighter, and the resulting effects were more pronounced.

4. Discussion

Figures 17–21 present the flow aspects around the model at a 0° roll angle, at incidences of 0° , 10° , 20° , 30° , 40° , and 50° for Mach numbers of 0.4, 0.95, 1.6, 2.5, and 3.5, aiming to understand the trends in aerodynamic coefficients as functions of both angle of attack and Mach number.

An initial observation is that the stagnation zone increased with angle of attack due to the model's length, leading to a significant increase in the normal force coefficient. Additionally, the appearance of normal shock waves in the transonic regime induced increased drag, while oblique, detached shockwaves in supersonic regimes lead to decreased drag compared to the transonic regime.

At zero incidence, the model's fins experienced the same incidence, leading to identical pressure distributions on their surfaces, resulting in null aerodynamic coefficients except for the rolling moment coefficient and axial force coefficient. The rolling moment was non-zero at zero incidence because the pressures on each fin's two surfaces differed due to their effective incidences, generating four force couples relative to the rolling axis, producing a moment. However, this rolling moment remained constant at zero incidence because the effective incidence on the fins remained constant regardless of the roll angle. This changed when the incidence increased, as the effective incidence on the fins varied with the roll angle, generating periodic variations in aerodynamic coefficients. The period of this variation was 90° , equal to the angular spacing between the fins.

Moreover, at non-zero incidences, both the normal force coefficient and lateral force coefficient derivative, as well as pitch moment coefficient and yaw moment coefficient derivative, presented non-zero values due to varying pressure distributions on the fin's surfaces. Dorsal fins presented lower pressure values due to interaction with flow detachment from the fuselage's surface, while ventral fins showed higher pressure values, as they reside in an undisturbed flow and may even present stagnation surfaces. Furthermore, in supersonic regimes, there was an interaction between the fins and the shock wave generated by the fuselage, leading to rapid changes in pressure distribution and substantial variations in aerodynamic coefficients at higher incidences. At high Mach numbers (Mach 2.5 and Mach 3.5) and large angles of attack (40° and 50°), the ventral bow shock lay very close to the fuselage, allowing the fins to interact with it and become partially exposed to the undisturbed flow, which in turn generated shock waves.

Aerodynamic contributions on the fuselage surface are also considerable, affecting not only the rolling moment coefficient, where friction generates a rolling moment, but also the normal or axial force coefficients, where flow detachment on the dorsal surface, suction on the model's base, or shock waves induced significant contributions.

Another notable observation involves the loss of efficiency in the dorsal fin at high incidences. Due to flow detachment from the fuselage's dorsal surface, fluid velocity significantly decreased, resulting in reduced pressure on the fin's surface. This interaction of the dorsal fin with flow detachment largely decreased the aerodynamic coefficients concerning the variation with the angle of attack, except for the axial force coefficient and rolling moment coefficient, which increased due to fin loading asymmetry. Conversely, at a 45° roll angle when the model was in the '×' position, dorsal fins tended to have high efficiency due to undisturbed flow interaction, except for interference in the fuselage's vicinity. Consequently, aerodynamic coefficients mostly tended to increase, except for the axial force coefficient and rolling moment coefficient.

The authors of [20] stated that aerodynamic spin derivatives are essential for lateral-directional stability and control analysis. Specifically, the roll mode is strongly influenced by the roll damping derivative, which directly affects the time constant T_r .

The roll mode time constant, T_r , quantifies the speed of the roll response. A smaller roll time constant indicates a quicker increase in roll rate following a lateral control input. The standard MIL-F-8785C specifies the maximum allowable roll mode time constant for various flight phases, aircraft classes, and stability levels, whereas missile stability and control standards are often classified.

The roll mode approximation of time constant is presented in [20] as: $T_r \approx -\frac{1}{L_p}$, where $L_p = \frac{qSb^2C_{lp}}{2I_{xx}V}$, q is the dynamic pressure, S is the reference area, b is the reference length, I_{xx} is the axial moment of inertia, V is the velocity, and C_{lp} is the roll moment coefficient. Therefore, a higher roll damping coefficient, C_{lp} , results in a smaller time constant, T_r , and improved stability.

5. Conclusions

This numerical study investigated the aerodynamic behavior of the Basic Finner Model in rolling motion, examining variation in aerodynamic coefficients with different Mach numbers (0.4, 0.95, 1.6, 2.5, and 3.5), angles of attack (0°, 10°, 20°, 30°, 40°, and 50°), and roll angles (0° to 360°). This study built open previous studies [2,4,19] and provides the most comprehensive numerical analysis of the Basic Finner Model in roll motion.

The study described the computational domain, discretization method, boundary conditions, solution models, and validation against experimental reference data, thereby providing a database characterizing the Basic Finner Model's aerodynamic roll damping.

The results reveal three derivative variations based on Mach number, incidence, and roll angle. Additionally, surface pressure coefficient (C_p) distributions on the model and Mach number distributions in the symmetry plane are presented for different incidences and Mach numbers. The purpose of these analyses was to characterize the Basic Finner Model's aerodynamic coefficients across a wide range of Mach numbers, incidence angles, and roll angles. The variations in aerodynamic coefficients with incidence and Mach number were strongly nonlinear, as a result of the complex flow behavior around the rotating model. This nonlinearity arose from factors such as shock wave interactions, flow separation, and dynamic changes in the wake, which were influenced by both Mach number and angle of attack. The derivatives varied periodically with roll angle, with a 90° period corresponding to the angular spacing between the fins.

To deepen understanding of derivative variation with roll angle, C_p and Mach number distributions in the symmetry plane were analyzed for different Mach numbers and incidences at a roll angle of 0° .

The C_p and Mach distributions at high incidences reveal strong interactions between the model's fins, the wake on the dorsal surface, and the ventral shock wave generated by the model body, particularly in supersonic regimes. These interactions correlated with derivative variations with roll angle, inducing abrupt variations with large amplitudes at high incidences, especially in supersonic regimes.

This study highlights several key findings. The roll damping coefficient was consistently negative, indicating roll-axis stability. However, it presented irregular variations with changes in both Mach number and angle of attack (AoA). The yaw damping coefficient showed positive values in the subsonic regime and negative values in the supersonic regime, with similar magnitudes. In the transonic regime, it remained negative up to an AoA of 30° , then became positive beyond that point. In contrast, the side force damping coefficient was negative in the subsonic regime and positive in the supersonic regime. For the transonic regime, it was positive up to 20° AoA and negative between 20° and 50° AoA. The pitch moment coefficient remained negative across all regimes and AoA values, confirming pitch-axis stability. In both subsonic and supersonic regimes, its magnitude decreased steadily with AoA. In the transonic regime, however, it decreased up to 20° AoA and then increased. The normal force coefficient was always positive and increased with AoA. Its rate of increase was lower in the subsonic regime compared to the steeper slopes observed in the transonic and supersonic regimes. The axial force coefficient was also consistently positive. It decreased with AoA in the subsonic and transonic regimes, increased slightly in the low supersonic regime, and showed a strong increasing trend in the high supersonic regime.

In conclusion, this numerical study provides a comprehensive database for the Basic Finner Model in rolling motion across a wide range of Mach numbers, angles of attack, and roll angles. It also deepens understanding of the flow physics around the rotating model, making it a valuable resource for further aerodynamic studies. The results obtained in this study can serve as a reference for future numerical or experimental investigations, given that the Basic Finner Model is a widely recognized standard for dynamic applications.

Author Contributions: Conceptualization, I.B., A.-I.O., T.-P.A. and M.-V.H.; methodology, I.B., A.-I.O. and T.-P.A.; software, M.-V.H.; validation, I.B., M.-G.S. and M.-V.P.; formal analysis, I.B.; investigation, I.B. and M.-V.P.; resources, M.-G.S.; data curation, M.-V.H.; writing—original draft preparation, I.B.; writing—review and editing, M.-G.S., A.-I.O., T.-P.A. and M.-V.P.; visualization, I.B.; supervision, M.-V.P.; project administration, M.-V.P.; funding acquisition, I.B. All authors have read and agreed to the published version of the manuscript.

Funding: This study was fully supported by the contract 671PED/ 21.06.2022 (PN-III-P2-2.1-PED-2021-2215), with the title “High Speed Aerodynamic Damping Identification Rigs for Roll and Pitch in Supersonic Wind Tunnel” (HiSAEROD), granted by the Executive Agency for Higher Education, Research, Development, and Innovation Funding (UEFISCDI).

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Acknowledgments: This work represents a relevant part of the first author's PhD thesis “Theoretical and experimental studies for determination of roll damping coefficient for cross configuration”, presented on 18 June 2024 in Bucharest, available on <https://rei.gov.ro/teze-doctorat>. This work was presented as Conference Paper AIAA 2024-2348 at the AIAA SciTech Forum, Orlando, FL, 8–12 January 2024.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AoA	Angle of attack [deg]
BFM	Basic Finner Model
SST	Shear stress transport

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